Engineering Chemistry

UNIT – I : Structure and Bonding Models

A clear, exam-ready walk through quantum fundamentals, the Schrödinger picture, and molecular-orbital bonding — with fully worked diagrams.

Topics in this unit

1. Fundamentals of quantum mechanics
2. Schrödinger wave equation; significance of Ψ and Ψ^2
3. Particle in a one-dimensional box
4. Molecular orbital theory (LCAO), σ & π bonds
5. Energy-level diagrams of O_2 and CO
6. π-MOs of butadiene & benzene; bond order

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. Why Quantum Mechanics? The Big Picture

Classical (Newtonian) mechanics works beautifully for cricket balls and planets, but it fails for tiny particles like electrons. Electrons are so small and fast that they behave partly like particles and partly like waves. To describe them we need a new rule-book — quantum mechanics — which treats the electron as a wave and tells us only the probability of finding it somewhere.

> NOTE
> This unit builds up in a logical chain: quantum basics → the Schrödinger equation → what Ψ means → a simple solved model (particle in a box) → and finally how all this explains real chemical bonding through Molecular Orbital Theory.

2. Fundamentals of Quantum Mechanics

Three key discoveries forced scientists to abandon the purely classical picture and accept that energy and matter are quantised (come in packets).

Planck's Quantum Theory (1900)

Energy is not emitted or absorbed continuously but in small discrete packets called quanta (a packet of light = a photon). The energy of one quantum is directly proportional to the frequency of radiation.

$$E = h\nu = hc / \lambda$$

h = Planck's constant (6.626×10^{-34} J·s), ν = frequency, λ = wavelength

Photoelectric Effect — Einstein (1905)

When light of high enough frequency strikes a metal, electrons are instantly ejected. This could only be explained if light behaves as particles (photons). Each photon gives its whole energy to one electron.

$$KE_{max} = h\nu - \varphi_0$$

φ_0 = work function (minimum energy to free an electron); below the threshold frequency ν_0, no electrons come out — however bright the light.

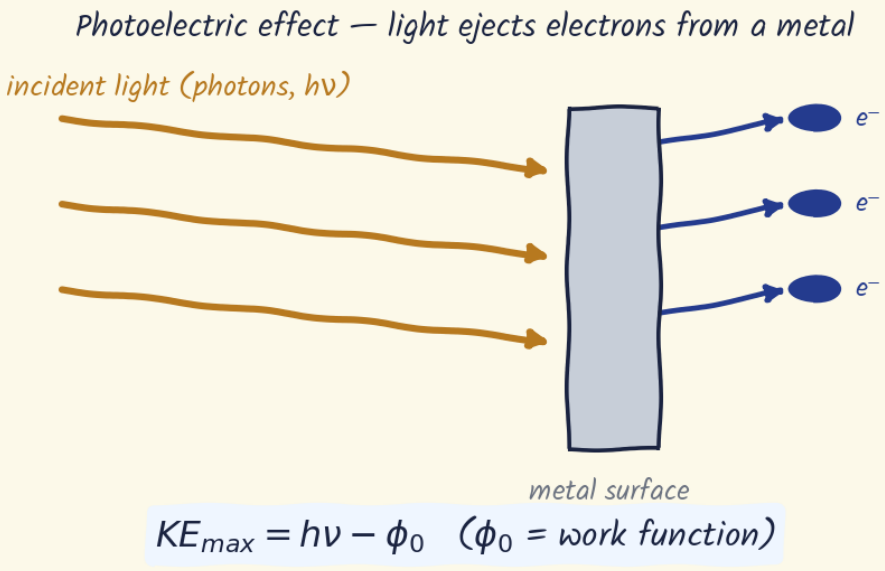


Fig 1 — Photons eject electrons from a metal surface; their kinetic energy depends on frequency, not intensity.

Wave-Particle Duality — de Broglie (1924)

If light (a wave) can behave like particles (photons), then matter (particles) can behave like waves. Every moving particle has a wavelength:

$$\lambda = h / mv = h / p$$

m = mass, v = velocity, p = momentum. Big objects → tiny λ (not noticeable); electrons → significant λ.

Heisenberg's Uncertainty Principle (1927)

It is impossible to know both the exact position and the exact momentum of an electron at the same time. The more precisely we fix one, the more uncertain the other becomes.

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

Δx = uncertainty in position, Δp = uncertainty in momentum

> KEY POINT
> Because we cannot pin-point an electron's path, the old idea of fixed Bohr orbits breaks down. Instead we speak of orbitals — regions of high probability of finding the electron.

3. Schrödinger Wave Equation (1926)

Erwin Schrödinger treated the electron as a standing wave around the nucleus and wrote a wave equation for it. Solving this equation gives the allowed energy levels of the electron and the wave functions (Ψ) that describe each state. It is the heart of modern atomic structure.

Time-independent form

$$\partial^2\Psi/\partial x^2 + \partial^2\Psi/\partial y^2 + \partial^2\Psi/\partial z^2 + (8\pi^2 m/h^2)(E - V)\Psi = 0$$

The compact operator form is: $\hat{H}\,\Psi = E\,\Psi$

What each term means

- Ψ (psi) — the wave function of the electron (what we solve for).
- $\hat{H}$ — the Hamiltonian operator (represents total energy: kinetic + potential).
- E — total energy of the electron (the allowed values we get out).
- V — potential energy of the electron; m — mass; h — Planck's constant.

NOTE

Only certain values of E give acceptable (well-behaved) solutions. This is exactly why energy is quantised — it drops out of the mathematics naturally.

4. Significance of Ψ and Ψ^2

This is a favourite exam question. The wave function Ψ and its square Ψ^2 mean very different things.

DEFINITION

Ψ (wave function): the amplitude of the electron wave. It can be positive, negative or zero and has no direct physical meaning by itself. It is just a mathematical function describing the electron's state.

DEFINITION

Ψ^2 (probability density): the probability of finding the electron in a unit volume around a point. It is always positive. A region where Ψ^2 is large is exactly what we call an orbital. (Born interpretation)

Conditions for a well-behaved wave function

- *Single-valued* — one value of Ψ at each point.
- *Finite* — Ψ must not become infinite anywhere.
- *Continuous* — Ψ and its slope change smoothly.
- *Normalised* — total probability over all space = 1.

KEY POINT

Memory hook: Ψ = address, Ψ^2 = how likely the electron is at that address. Only Ψ^2 can be measured; Ψ on its own cannot.

5. Particle in a One-Dimensional Box

This is the simplest problem we can solve exactly with the Schrödinger equation, and it shows why energy is quantised in a very clear way.

The model

- A particle of mass m is trapped in a 1-D box of length a.
- Inside the box the potential energy $V = 0$ (particle moves freely).
- At and beyond the walls $V = \infty$, so the particle can never escape.

Boundary conditions & solution

Since the particle cannot exist at the walls, $\Psi = 0$ at $x = 0$ and $x = a$.

Applying these to the Schrödinger equation gives the allowed wave functions and energies:

$$\Psi_n = \sqrt{(2/a)} \cdot \sin(n\pi x/a) \quad E_n = n^2h^2 / 8ma^2$$

$n = 1, 2, 3, \ldots$ is the quantum number

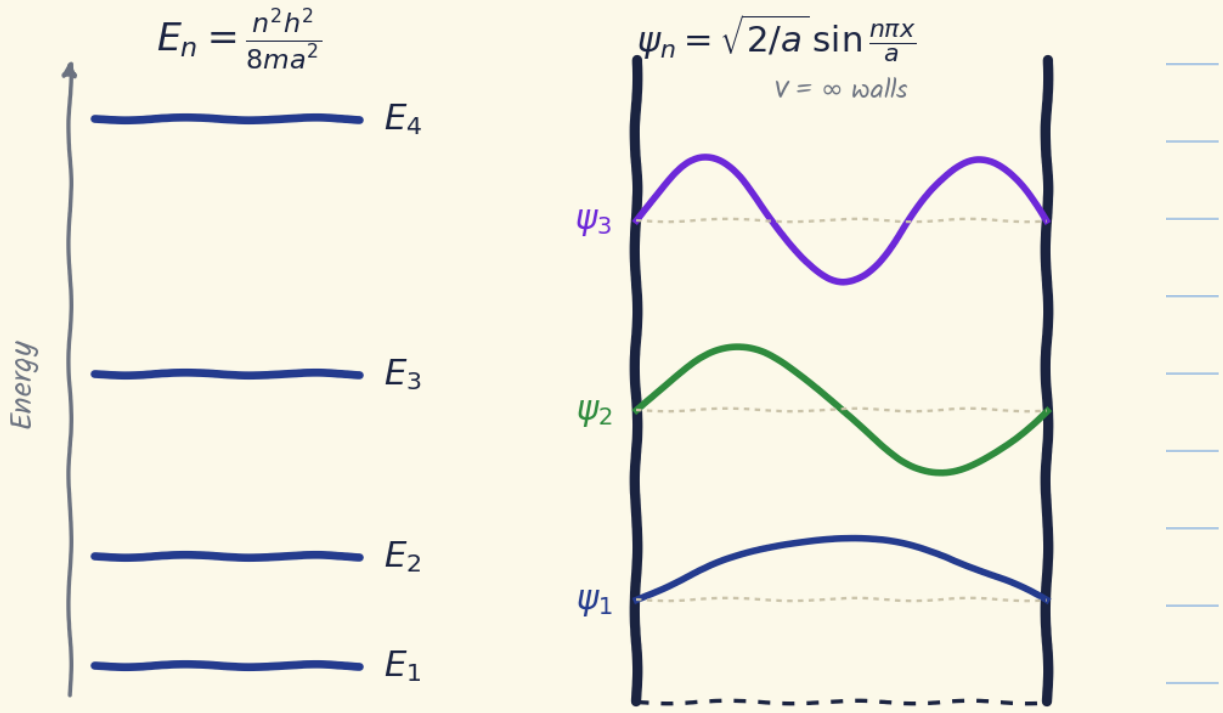


Fig 1 — Quantised energy levels (left) and the first three wave functions inside the box (right).

What this tells us

- **Energy is quantised** — only $n = 1, 2, 3\ldots$ are allowed, never values in between.
- **Zero-point energy:** the lowest level $E_1 = h^2/8ma^2$ is **not zero** — a trapped particle can never be completely at rest.
- Energy levels get **farther apart** as n increases ($E \propto n^2$): ratio $1 : 4 : 9 : 16\ldots$
- Number of nodes (points where $\psi = 0$ inside) = $n - 1$.

> **EXAMPLE**
> For $n = 1, 2, 3, 4$ the energies are in the simple ratio $1 : 4 : 9 : 16$, because $E_n \propto n^2$. So E_2 is $4\times$ E_1.

6. Molecular Orbital Theory (MOT)

Proposed by Hund and Mulliken, MOT is the most successful model of covalent bonding. Its central idea: when atoms bond, their atomic orbitals (AOs) merge and lose their individual identity, forming new molecular orbitals (MOs) that belong to the whole molecule.

LCAO — how MOs form

MOs are built by the Linear Combination of Atomic Orbitals (LCAO).

Combining n atomic orbitals always gives n molecular orbitals:

- Addition (constructive overlap, waves in phase) → bonding MO, lower energy, more stable. Example: σ, π.
- Subtraction (destructive overlap, waves out of phase) → antibonding MO (marked with *), higher energy, less stable. Example: σ*, π*.

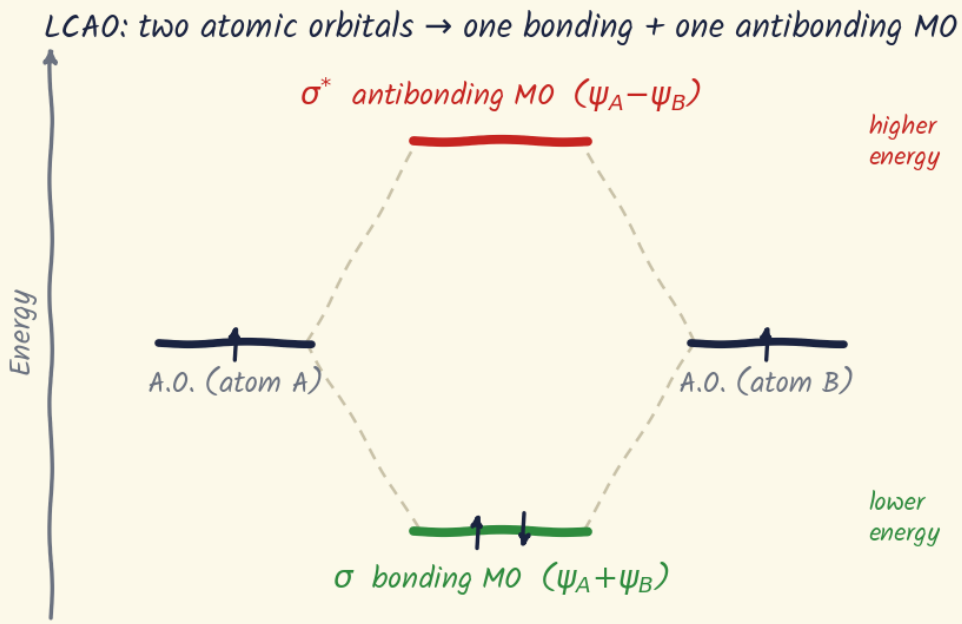


Fig 2 — Two atomic orbitals combine into one bonding (lower) and one antibonding (higher) molecular orbital.

σ and π molecular orbitals

The way orbitals overlap decides the type of MO: end-on overlap gives σ bonds (electron cloud along the bond axis) while sideways overlap gives π bonds (cloud above and below the axis).

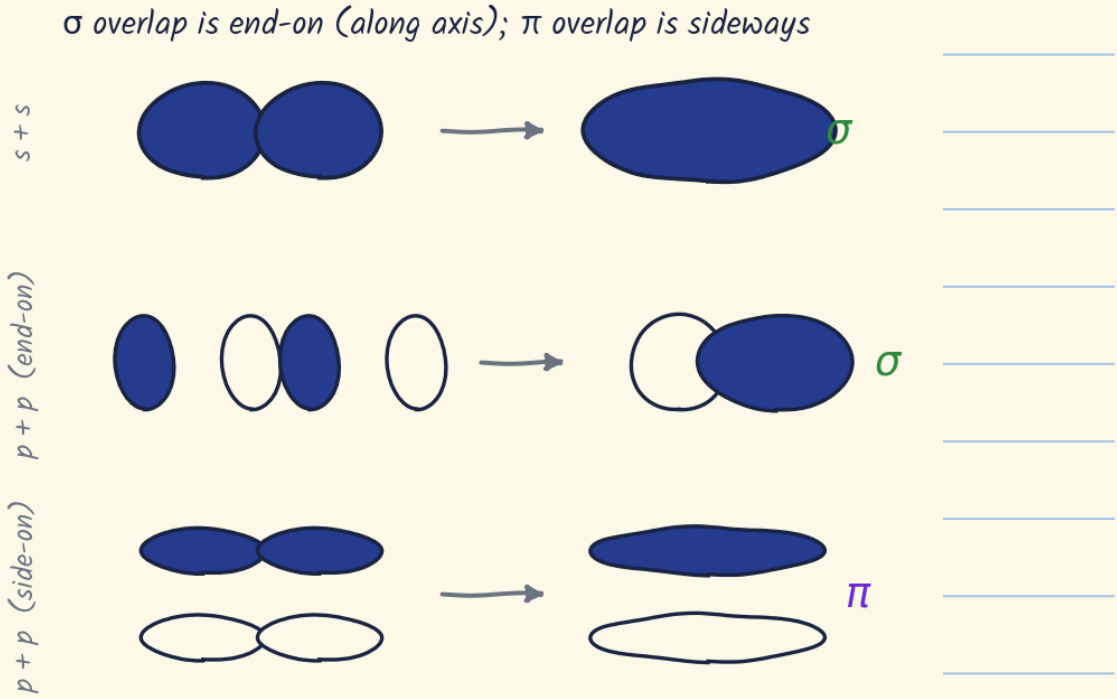


Fig 3 — σ overlap is end-on; π overlap is sideways.

Rules for filling MOs

Electrons fill molecular orbitals using the same three rules as atoms: Aufbau (lowest energy first), Pauli (max 2 e per MO, opposite spins) and Hund's rule (degenerate MOs are singly filled first).

Energy order of molecular orbitals

Up to N_2: $\sigma 1s\ \sigma^*1s\ \sigma 2s\ \sigma^*2s\ (\pi 2p_x = \pi 2p_y)\ \sigma 2p_z\ (\pi^*2p_x = \pi^*2p_y)\ \sigma^*2p_z$

For O_2, F_2 and beyond, σ2pz drops BELOW the π2p pair (no s–p mixing)

NOTE

Why two orders? For lighter molecules (B_2, C_2, N_2) s–p mixing pushes $\sigma 2p_z$ above the $\pi 2p$ orbitals. From O_2 onward this mixing is weak, so $\sigma 2p_z$ sits below $\pi 2p$. This single difference explains O_2's magnetism — see below.

7. Bond Order

Bond order is a quick number that tells us how strong and stable a bond is.

DEFINITION

Bond order (B.O.): half the difference between the number of electrons in bonding MOs (N_b) and antibonding MOs (N_a).

$$\text{Bond order} = \tfrac{1}{2}\,(N_b - N_a)$$

- B.O. > 0 → molecule is stable; B.O. = 0 → molecule does not exist.
- Higher bond order → stronger bond, shorter bond length, higher bond energy.
- Unpaired electrons → paramagnetic; all paired → diamagnetic.

8. Energy-Level Diagram of O_2

Oxygen atom ($Z = 8$): $1s^2\ 2s^2\ 2p^4$. The O_2 molecule therefore has 16 electrons, filled into MOs using the O_2-type order ($\sigma 2p_z$ below $\pi 2p$).

$$O_2:\ KK\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^{\,2}\ (\pi 2p_x^{\,2}\ \pi 2p_y^{\,2})\ (\pi^*2p_x^{\,1}\ \pi^*2p_y^{\,1})$$

KK = filled 1s MOs ($\sigma 1s^2\ \sigma^*1s^2$)

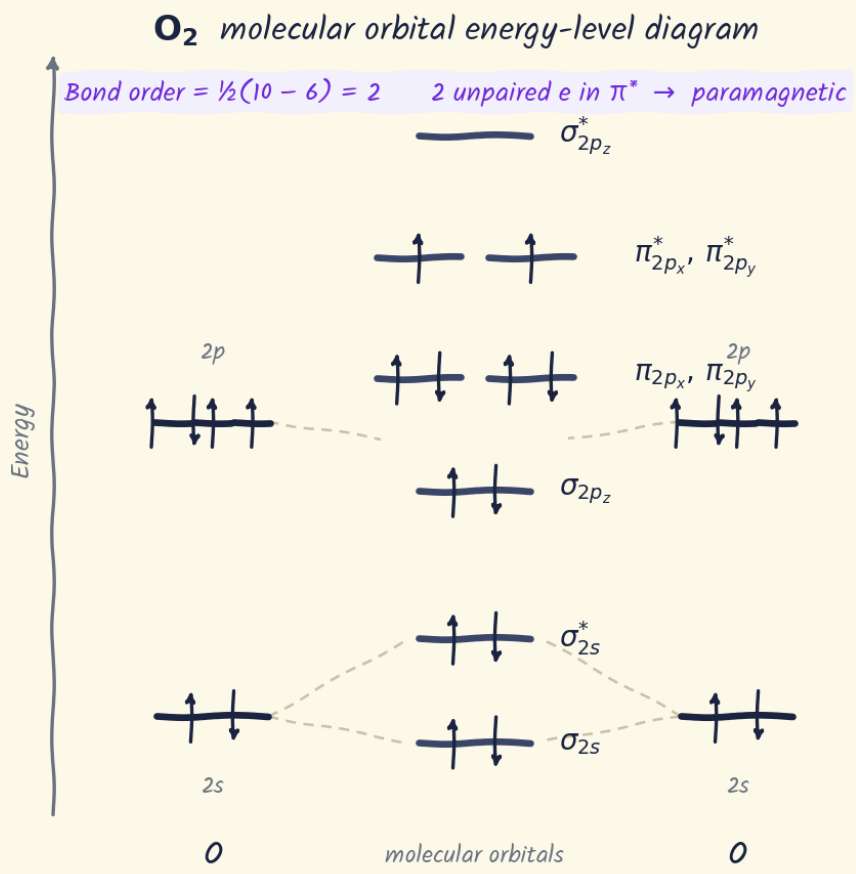


Fig 4 — Molecular-orbital energy-level diagram of O_2. The two π^* orbitals each hold one unpaired electron.

Results

- $N_b = 10$, $N_a = 6$ → Bond order = ½(10 − 6) = 2 (a double bond, O=O).
- Two unpaired electrons sit in the π^* orbitals → O_2 is paramagnetic.

> KEY POINT
>
> This is the biggest success of MOT. Lewis structures and Valence Bond Theory wrongly predict O_2 to be diamagnetic; only MOT correctly explains why liquid oxygen is attracted by a magnet.

9. Energy-Level Diagram of CO

Carbon ($Z = 6$): $1s^2\ 2s^2\ 2p^2$; Oxygen ($Z = 8$): $1s^2\ 2s^2\ 2p^4$. CO has 14 electrons and is isoelectronic with N_2, so it uses the N_2-type order ($\pi 2p$

below $\sigma 2p_z$).

$$CO:\ KK\ \sigma 2s^2\ \sigma^*2s^2\ (\pi 2p_x^{\,2}\ \pi 2p_y^{\,2})\ \sigma 2p_z^{\,2}$$

π^* and σ^*2pz remain empty

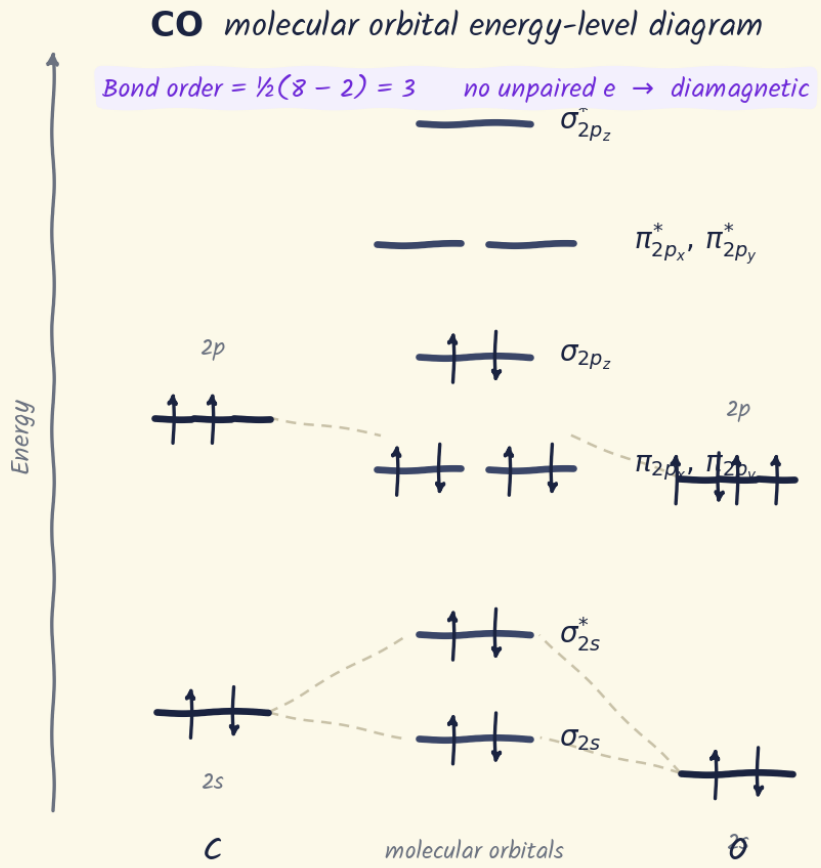


Fig 5 — MO energy-level diagram of CO (O orbitals lie a little lower as O is more electronegative).

Results

- $N_b = 10$, $N_a = 4$ → Bond order = ½(10 − 4) = 3 (a triple bond, C≡O).
- No unpaired electrons → CO is diamagnetic.
- The high bond order explains CO's very high bond energy and short bond length.

10. π-Molecular Orbitals of Butadiene

1,3-Butadiene ($CH_2=CH-CH=CH_2$) has four carbon atoms in a row, each giving one p-orbital. These 4 p-orbitals combine into 4 π molecular orbitals (ψ_1, ψ_2, ψ_3, ψ_4).

- The number of nodes increases with energy: ψ_1 has 0, ψ_2 has 1, ψ_3 has 2, ψ_4 has 3.
- There are 4 π electrons; they fill the two lowest MOs — ψ_1 and ψ_2.
- ψ_2 = HOMO (highest occupied), ψ_3 = LUMO (lowest unoccupied).

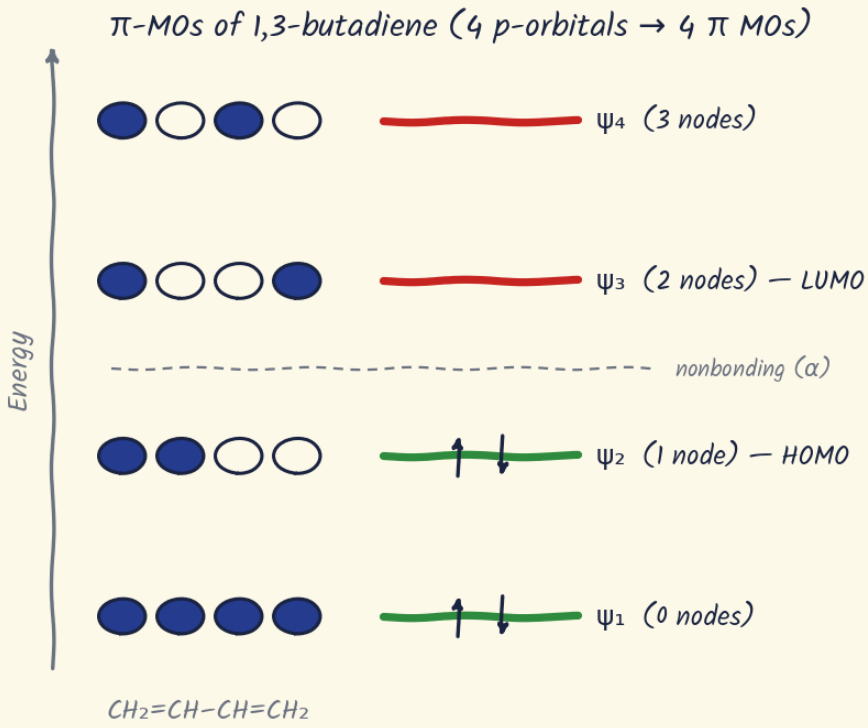


Fig 6 — The four π MOs of butadiene with their node patterns and electron filling.

> KEY POINT
> Because the π electrons are delocalised over all four carbons (not fixed between two atoms), butadiene is more stable than two isolated double bonds. This extra stability is the resonance / conjugation energy.

II. π-Molecular Orbitals of Benzene

Benzene (C_6H_6) has six carbons in a ring, each contributing a p-orbital. The 6 p-orbitals combine into 6 π MOs: three bonding (ψ_1, ψ_2, ψ_3) and three antibonding (ψ_4^*, ψ_5^*, ψ_6^*).

- ψ_1 is lowest; ψ_2 and ψ_3 are a degenerate pair (equal energy), as are ψ_4^* and ψ_5^*.
- Benzene's 6 π electrons exactly fill the 3 bonding MOs — a completely filled, very stable set.
- The result is a delocalised π cloud above and below the ring.

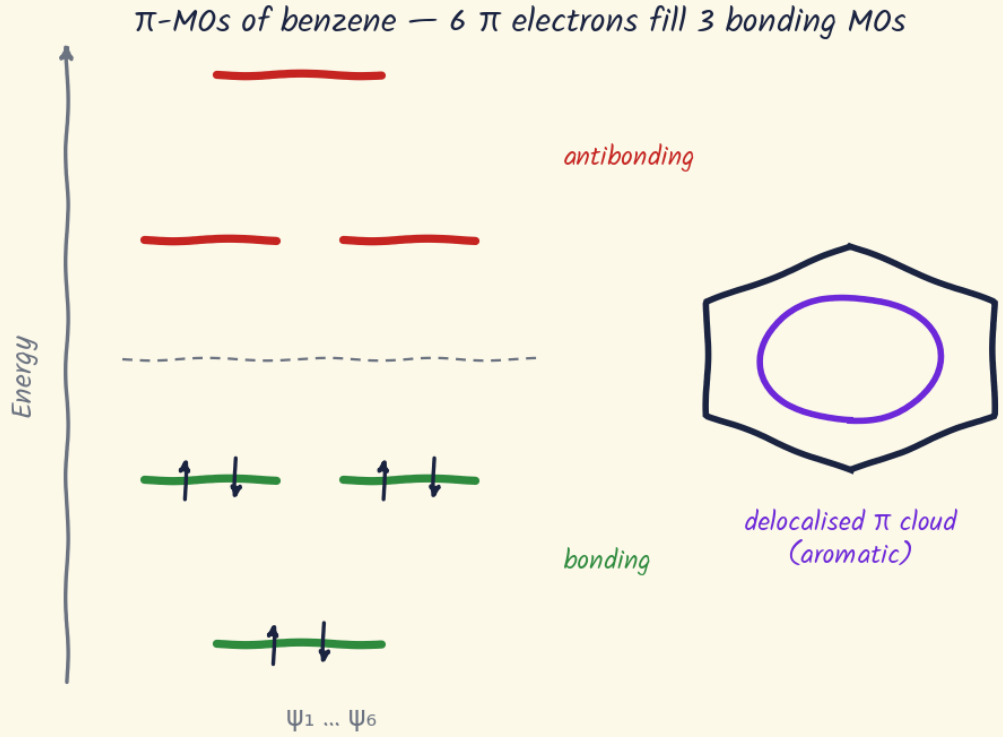


Fig 7 — Energy order of benzene's six π MOs and the delocalised π cloud over the ring.

KEY POINT

This full set of bonding MOs gives benzene its exceptional stability — its aromaticity (Hückel's rule: $4n+2$ π electrons, here $n = 1 \rightarrow 6$ electrons). This is why benzene resists addition reactions and prefers substitution.

12. Bond Orders at a Glance

Using B.O. = ½(N_b − N_a), here are common diatomic molecules. Notice how bond order tracks bond strength, and unpaired electrons decide magnetism.

Species	N_b	N_a	Bond order	Nature
H_2	2	0	1 (single)	Diamagnetic
N_2	10	4	3 (triple)	Diamagnetic
O_2	10	6	2 (double)	Paramagnetic
F_2	10	8	1 (single)	Diamagnetic
CO	10	4	3 (triple)	Diamagnetic
NO	10	5	2.5	Paramagnetic

EXAMPLE

He_2 would have N_b = 2 and N_a = 2, giving B.O. = 0. That is why the He_2 molecule does not exist — MOT predicts it cleanly.

13. Applications & Importance

- Magnetic behaviour: explains paramagnetism of O_2, B_2 and NO (unpaired e^-).
- Stability & existence: bond order tells whether a molecule/ion is stable (He_2, Be_2 with B.O. = 0 do not exist).
- Bond length & energy: higher bond order → shorter, stronger bonds.

- **Delocalisation & aromaticity:** explains the stability of butadiene, benzene and conjugated dyes.
- **Foundation for materials:** the band theory of metals, semiconductors and conductors (Unit II) grows directly out of MO ideas.

QUICK RECAP

- Quantum mechanics is needed because electrons show wave–particle duality and obey Heisenberg's uncertainty principle.
- The Schrödinger equation ($\hat{H}\Psi = E\Psi$) gives quantised energies and wave functions Ψ.
- Ψ has no physical meaning; Ψ^2 is the probability density (an orbital).
- Particle in a box: $E_n = n^2h^2/8ma^2$ — energy is quantised with a non-zero ground state (zero-point energy).
- MOT: AOs combine (LCAO) into bonding and antibonding MOs; Bond order = $\frac{1}{2}(N_b - N_a)$.
- O_2 is paramagnetic (2 unpaired e^-, B.O. 2); CO has a triple bond (B.O. 3, diamagnetic).
- Delocalised π-MOs explain the stability of butadiene and the aromaticity of benzene.

EXAM-ORIENTED QUESTIONS

Q1. State and explain Heisenberg's uncertainty principle. *[3 marks]*

Ans. It is impossible to measure the exact position and exact momentum of an electron simultaneously: $\Delta x \cdot \Delta p \geq h/4\pi$. Fixing one more precisely makes the other more uncertain — which is why electrons are described by orbitals, not fixed orbits.

Q2. What is the physical significance of Ψ and Ψ²? [3 marks]

Ans. Ψ is the wave function (amplitude of the electron wave) and has no direct physical meaning. Ψ^2 is the probability density — the chance of finding the electron in a unit volume; it is always positive and defines the orbital.

Q3. Write the Schrödinger wave equation and name its terms. [5 marks]

Ans. $\hat{H}\Psi = E\Psi$, i.e. $\partial^2\Psi/\partial x^2 + \partial^2\Psi/\partial y^2 + \partial^2\Psi/\partial z^2 + (8\pi^2 m/h^2)(E - V)\Psi = 0$, where Ψ = wave function, E = total energy, V = potential energy, m = mass, h = Planck's constant.

Q4. Discuss the particle in a 1-D box. What is zero-point energy? [5 marks]

Ans. A particle in a box of length a (V = 0 inside, ∞ at walls) has $E_n = n^2h^2/8ma^2$. Energy is quantised. The lowest level $E_1 = h^2/8ma^2$ is non-zero — this minimum energy is the zero-point energy, meaning the particle is never fully at rest.

Q5. Distinguish between bonding and antibonding molecular orbitals. [3 marks]

Ans. Bonding MOs form by constructive (in-phase) overlap, have lower energy and increase stability (σ, π). Antibonding MOs form by destructive (out-of-phase) overlap, have higher energy, carry a node between nuclei and reduce stability (σ^*, π^*).

Q6. Draw the MO diagram of O_2 and explain its paramagnetism. [5 marks]

Ans. O_2 (16 e^-): KK $\sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p^4\ \pi^*2p^2$. The two π^* electrons remain unpaired (Hund's rule), making O_2 paramagnetic; B.O. = ½(10−6) = 2.

Q7. Calculate the bond order of CO and give its magnetic nature. [3 marks]

Ans. CO (14 e^-, N_2-type order): $N_b = 10$, $N_a = 4$, so B.O. = ½(10−4) = 3 (triple bond). All electrons are paired → diamagnetic.

Q8. Why does the He_2 molecule not exist? [2 marks]

Ans. In He_2, $N_b = 2$ and $N_a = 2$, giving bond order 0. With no net bonding, He_2 is not a stable molecule.

Q9. Explain the π-MOs and aromaticity of benzene. [3 marks]

Ans. Benzene's 6 p-orbitals give 6 π MOs (3 bonding, 3 antibonding). Its 6 π electrons fully fill the 3 bonding MOs, creating a stable delocalised π cloud — this is aromaticity (Hückel 4n+2 rule, n = 1).

Q10. State the de Broglie relation and its significance. [2 marks]

Ans. $\lambda = h/mv$. Every moving particle has an associated wavelength; it is significant for light, fast particles like electrons and is the basis of their wave behaviour.

Q11. Define an orbital. How is it different from a Bohr orbit? [3 marks]

Ans. An orbital is a region of space where the probability (Ψ^2) of finding an electron is high. Unlike a fixed Bohr orbit (a definite circular path), an orbital gives only a probability, which agrees with Heisenberg's uncertainty principle.

Q12. What is LCAO? [2 marks]

Ans. Linear Combination of Atomic Orbitals — atomic orbitals are added and subtracted to build molecular orbitals. Combining n atomic orbitals always gives n molecular orbitals (bonding + antibonding).

Q13. Why does the N_2-type MO order differ from the O_2-type order? [3 marks]

Ans. In lighter molecules (up to N_2), s–p mixing pushes $\sigma 2p_z$ above the $\pi 2p$ pair. From O_2 onward this mixing is weak, so $\sigma 2p_z$ lies below $\pi 2p$. This is why O_2 ends up with unpaired π^* electrons (paramagnetic).

Q14. Compare σ and π bonds. [3 marks]

Ans. A σ bond forms by end-on (axial) overlap and is stronger, with electron density along the bond axis. A π bond forms by sideways overlap of p-orbitals and is weaker, with electron density above and below the axis.

Q15. Calculate the bond order of N_2 and give its magnetic nature. [3 marks]

Ans. N_2 (14 electrons): $N_b = 10$, $N_a = 4$, so B.O. = $\frac{1}{2}(10 - 4) = 3$ (a triple bond). All electrons are paired, so N_2 is diamagnetic.

Q16. State Planck's quantum theory. [2 marks]

Ans. Energy is emitted or absorbed only in small discrete packets called quanta, with $E = h\nu$ (proportional to frequency) — not continuously.

Q17. What is a node in a wave function? [2 marks]

Ans. A node is a point or region where $\Psi = 0$, so the probability of finding the electron there is zero. For a particle in a box the number of nodes = $n - 1$.

Engineering Chemistry

UNIT – 2 : Modern Engineering Materials

A clear, exam-ready tour of the materials behind modern technology — semiconductors, superconductors, supercapacitors and the carbon nanoworld — with hand-drawn diagrams.

Topics in this unit

1. Semiconductors — concept & applications
2. Superconductors — concept & applications
3. Supercapacitors — concept, classification & applications
4. Nanomaterials — classification & properties
5. Fullerenes, carbon nanotubes & graphene

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. Modern Engineering Materials — The Big Picture

Behind every new technology — a faster phone chip, an MRI scanner, an electric-car battery, a lighter aircraft — there is a special material making it possible. This unit introduces five families of modern materials and what makes each one remarkable.

NOTE

Our journey: semiconductors (the heart of all electronics) → superconductors (zero resistance) → supercapacitors (super-fast energy storage) → nanomaterials (matter at the billionth-of-a-metre scale) → the carbon wonder-trio fullerenes, carbon nanotubes & graphene. One idea links them all: how freely electrons can move, set by a material's energy bands — so that is where we begin.

2. Semiconductors & Band Theory

A semiconductor is a material whose ability to conduct electricity lies between a conductor and an insulator. Silicon (Si) and germanium (Ge) are the famous examples — they run every computer chip in the world.

DEFINITION

Semiconductor: a solid whose electrical conductivity is between that of a conductor and an insulator, controlled by a small energy band gap and strongly affected by temperature and doping.

Band theory of solids

In a solid, millions of atomic energy levels merge into continuous energy bands. Two of them matter most:

- Valence band (VB) — the lower band, normally full of electrons.
- Conduction band (CB) — the upper band, normally empty; electrons here are free to move and carry current.
- Band gap (E_g) — the forbidden energy region between VB and CB. Its width decides how the solid behaves.

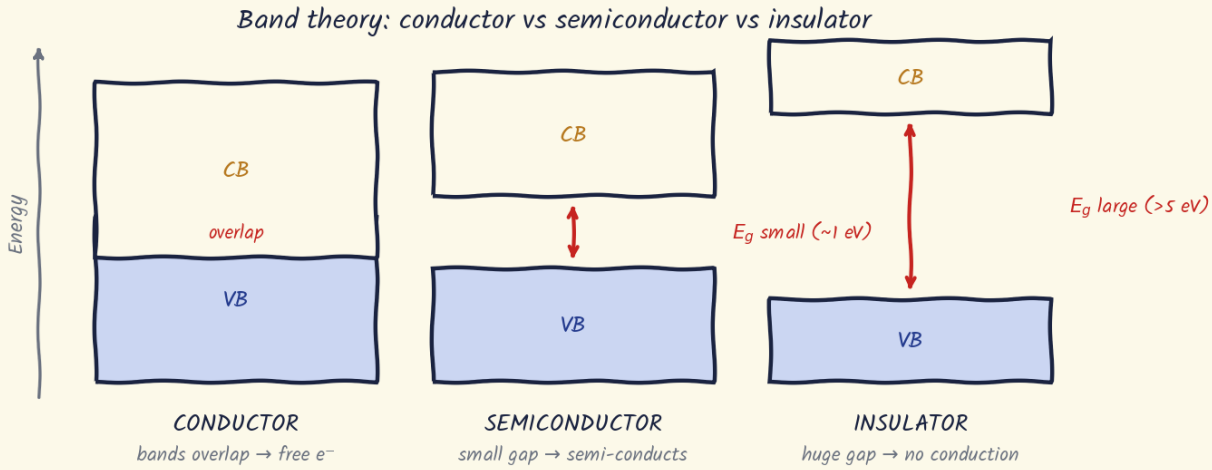


Fig 1 — Conductors have overlapping bands (no gap), semiconductors a small gap, insulators a large gap.

Conductor vs semiconductor vs insulator

Type	Band gap E_g	Example	Conducts?
Conductor	0 (bands overlap)	Cu, Al, metals	yes, freely
Semiconductor	small (~0.5–3 eV)	Si (1.1 eV), Ge (0.7 eV)	a little; more when hot
Insulator	large (>5 eV)	diamond	no

KEY POINT

Unlike metals, a semiconductor's conductivity increases as temperature rises. Heat gives more electrons enough energy to jump the small gap into the conduction band, so more charge carriers appear. (At 0 K a pure semiconductor behaves like an insulator — the idea behind thermistors.)

3. Intrinsic & Extrinsic Semiconductors

Semiconductors come in two kinds: pure ones and deliberately impure ones. Adding a tiny, controlled amount of impurity is the trick that makes all modern electronics possible.

Intrinsic (pure) semiconductors

DEFINITION

Intrinsic semiconductor: a pure semiconductor (just Si or Ge) with no added impurity. Its few carriers come only from electrons thermally excited across the gap, leaving behind an equal number of holes.

- Equal numbers of electrons (in CB) and holes (in VB).
- Very low conductivity at room temperature.

Extrinsic (doped) semiconductors

DEFINITION

Doping: adding a tiny amount of impurity (about 1 atom in a million) to a pure semiconductor to greatly raise its conductivity. The product is an extrinsic semiconductor.

n-type — doped with group 15

Add a pentavalent impurity (P, As — group 15). Four of its five valence electrons form bonds; the fifth is left over and almost free.

- The impurity is a donor — it sits as a donor level just below the conduction band and easily gives up its spare electron.
- Majority carriers = electrons (negative) → hence n-type.

p-type — doped with group 13

Add a trivalent impurity (B, Al — group 13). With only three valence electrons, one bond is left incomplete — a hole.

- The impurity is an acceptor — it sits as an acceptor level just above the valence band and readily accepts an electron.
- Majority carriers = holes (positive) → hence p-type.

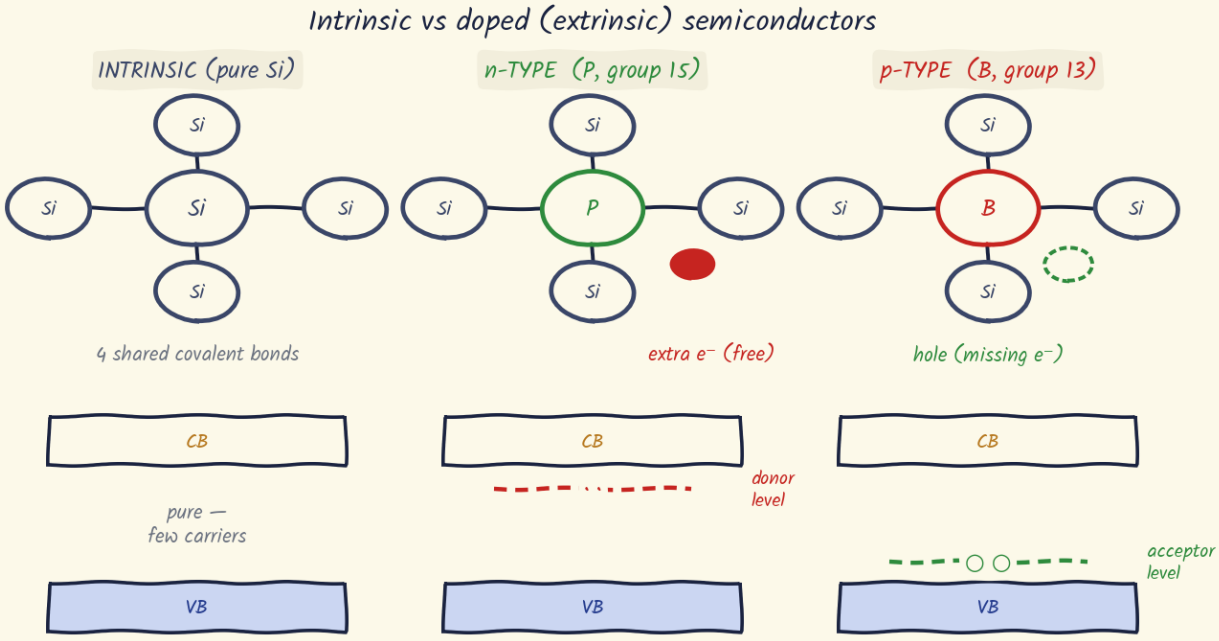


Fig 2 — Pure Si vs n-type (P donor, spare electron) vs p-type (B acceptor, hole), showing the donor and acceptor energy levels.

Feature	n-type	p-type
Dopant	group 15 (P, As) — pentavalent	group 13 (B, Al) — trivalent

Feature	n-type	p-type
Extra species	spare electron	hole
Majority carrier	electrons (−)	holes (+)
Impurity acts as	donor (level below CB)	acceptor (level above VB)

EXAMPLE

Joining a p-type piece to an n-type piece makes a p–n junction — the basis of the diode, the transistor and the solar cell, which are the building blocks of every chip.

Applications of semiconductors

- Diodes & rectifiers — convert AC to DC.
- Transistors & ICs (microchips) — switching and amplification in all computers.
- Solar (photovoltaic) cells — turn sunlight into electricity.
- LEDs, photodiodes & lasers — emitting and sensing light.
- Sensors — for temperature, light, pressure and gas.

4. Superconductors

Imagine a wire that carries current forever with no energy loss. That is a superconductor — one of the most striking effects in all of physics.

DEFINITION

Superconductor: a material that shows exactly zero electrical resistance when cooled below a certain critical temperature (T_c).

The effect was discovered by Kamerlingh Onnes in 1911, who cooled mercury and watched its resistance vanish at $T_c = 4.2$ K.

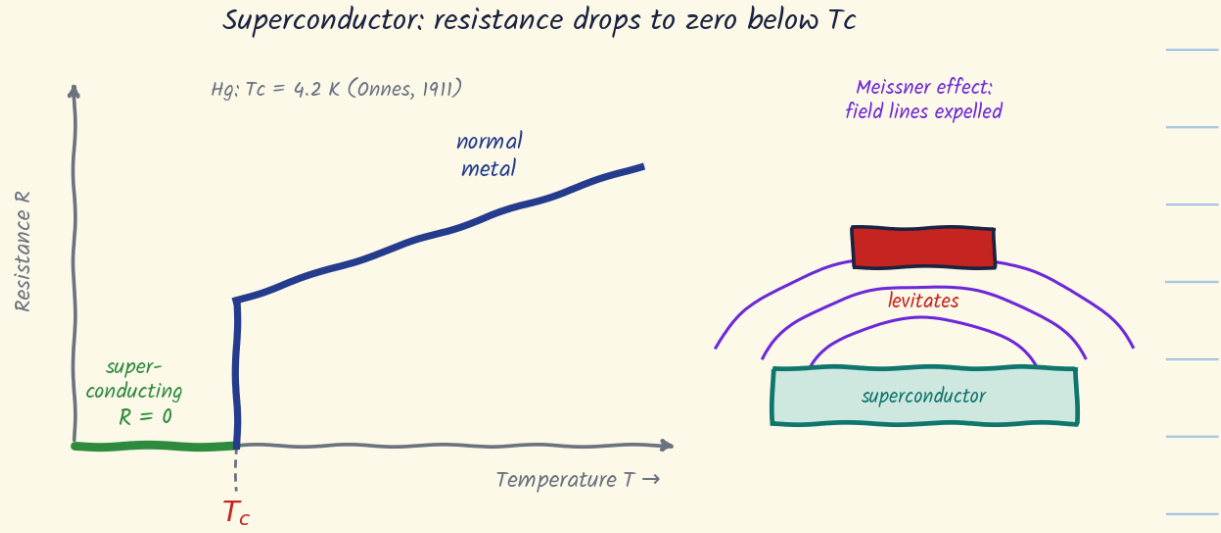


Fig 3 — Resistance drops sharply to zero at T_c; below T_c a magnet floats above the superconductor (Meissner effect).

Meissner effect

DEFINITION

Meissner effect: below T_c a superconductor completely expels magnetic field lines from its interior (perfect diamagnetism). This is why a magnet levitates above it.

KEY POINT

The Meissner effect is the true test of superconductivity and is the basis of magnetic levitation (Maglev) trains.

Type I and Type II

Feature	Type I (soft)	Type II (hard)
Made of	pure metals	alloys & ceramics

Feature	Type I (soft)	Type II (hard)
T_c	very low	higher
Magnetic field	expels fully, then breaks down	allows partial entry; stays super to high fields
Main use	little practical use	powerful magnets (MRI, Maglev)

A famous high-T_c ceramic is $YBa_2Cu_3O_7$ (YBCO), which superconducts at $T_c \approx 92$ K — above the boiling point of cheap liquid nitrogen (77 K).

BCS theory — why it happens

DEFINITION

BCS theory: below T_c the electrons pair up into **Cooper pairs** that move through the crystal lattice together **without scattering** — so there is no resistance.

Applications

- **MRI scanners** — superconducting magnets create the strong magnetic field.
- **Maglev trains** — frictionless magnetic levitation.
- **SQUIDs** — ultra-sensitive magnetometers.
- **Lossless power transmission** and very powerful electromagnets.
- **Particle accelerators** (e.g. the LHC).

5. Supercapacitors

A **supercapacitor** (or ultracapacitor) stores far more energy than an ordinary capacitor and charges far faster than a battery — it **bridges**

the gap between the two.

DEFINITION

Supercapacitor: an electrochemical energy-storage device that stores charge mostly physically (electrostatically) on very high-surface-area electrodes, giving very fast charge-discharge and a very long life.

Key properties

- Very fast charge and discharge (seconds).
- Very long cycle life — over 100,000 cycles.
- High power density, but lower energy density than a battery.

Property	Capacitor	Supercapacitor	Battery
Charge storage	electrostatic	mostly electrostatic	chemical
Charge speed	very fast	fast (seconds)	slow
Energy stored	very low	medium	high
Cycle life	huge	very long ($>10^5$)	limited

Classification

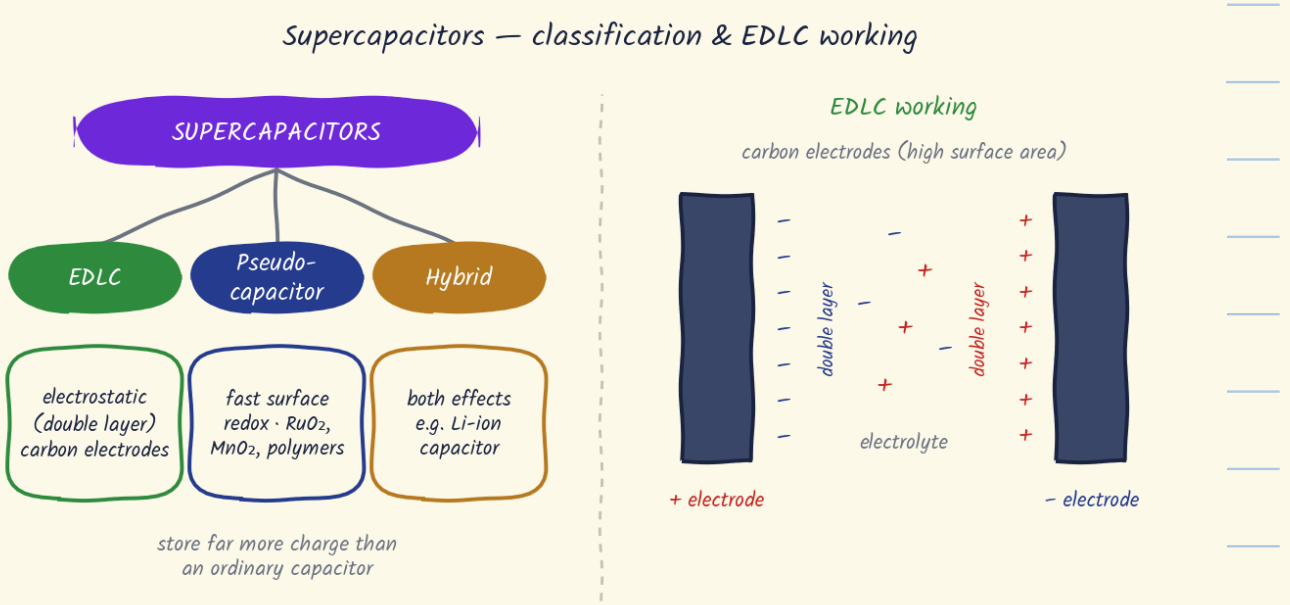


Fig 4 — The three types of supercapacitor and how an EDLC stores charge in the electric double layer.

1. EDLC — Electric Double-Layer Capacitor

Charge is stored electrostatically in the double layer that forms at the interface between a high-surface-area carbon electrode and the electrolyte. There is no chemical reaction — it is purely physical.

2. Pseudocapacitor

Stores charge through fast, reversible surface redox reactions (faradaic). It uses metal oxides (RuO_2, MnO_2) or conducting polymers.

3. Hybrid

Combines both mechanisms (electrostatic + redox) to store more energy. A common example is the lithium-ion capacitor.

Applications

- Regenerative braking in electric vehicles and trains.
- Memory and backup power and UPS systems.
- Camera flash and quick burst-power.

- Hybrid vehicles and buffering renewable energy (solar, wind).

6. Nanomaterials

A nanometre (nm) is one-billionth of a metre — about the width of a few atoms. Materials built at this scale behave in surprising new ways.

> DEFINITION
> Nanomaterial: a material with at least one dimension in the 1–100 nm range. At this size its properties differ greatly from the same material in bulk.

Why are nanomaterials special?

- Huge surface-area-to-volume ratio — most atoms sit on the surface, so the material is far more reactive and makes an excellent catalyst.
- Quantum confinement — electrons squeezed into tiny spaces change the optical, electrical and magnetic behaviour (e.g. colour can depend on particle size).

Classification by dimension

Nanomaterials are grouped by how many of their dimensions lie outside the nanoscale — giving 0-D, 1-D, 2-D and 3-D types.

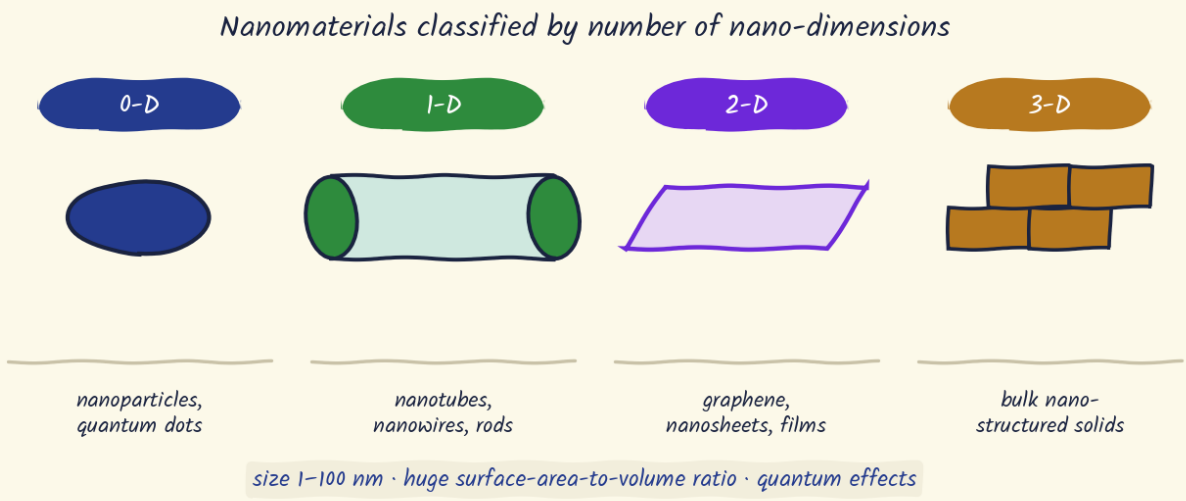


Fig 5 — Nanomaterials classified as 0-D, 1-D, 2-D and 3-D by the number of nano-sized dimensions.

- 0-D — every dimension nanoscale: nanoparticles, quantum dots.
- 1-D — one dimension larger: nanotubes, nanowires, nanorods.
- 2-D — two dimensions larger: graphene, nanosheets, thin films.
- 3-D — bulk nanostructured solids built from nanoscale grains.

KEY POINT

Same material, new powers: bulk gold is inert and yellow, but gold nanoparticles are brightly coloured and chemically active. At the nanoscale, size itself becomes a property.

7. Fullerenes — C_{60}

The carbon wonder-trio begins with the fullerene — a football-shaped molecule of pure carbon, discovered in 1985.

DEFINITION

Fullerene (C_{60}): a closed, hollow cage of 60 carbon atoms arranged as 20 hexagons + 12 pentagons — exactly like a soccer ball. Its full name is Buckminsterfullerene.

Structure & properties

- 60 sp^2 carbon atoms forming a spherical cage.
- Hollow and remarkably stable.
- A good electron acceptor (takes up electrons easily).

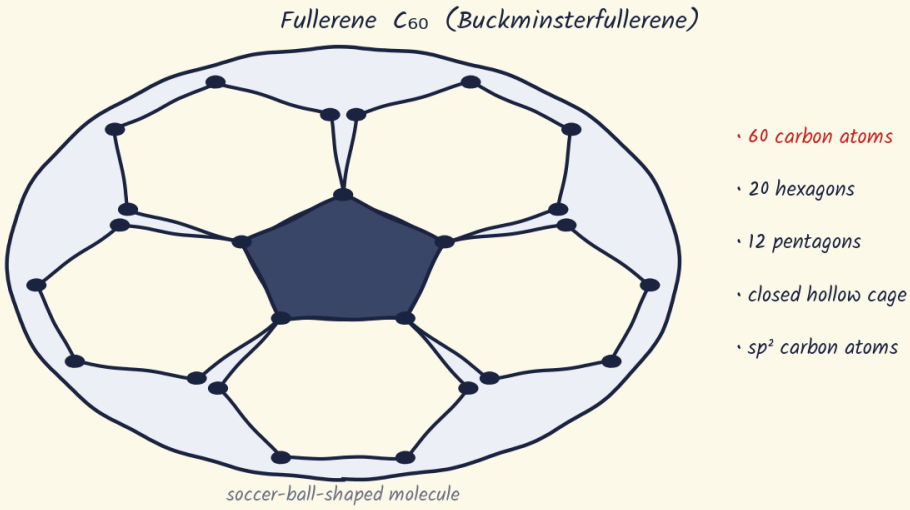


Fig 6 — C_{60}: a closed cage of 60 carbons — 20 hexagons and 12 pentagons — shaped like a soccer ball.

Applications

- Drug delivery — molecules carried inside the cage.
- Lubricants — act like tiny ball-bearings.
- Catalysts and organic solar cells.
- Superconductors (when doped) and antioxidants.

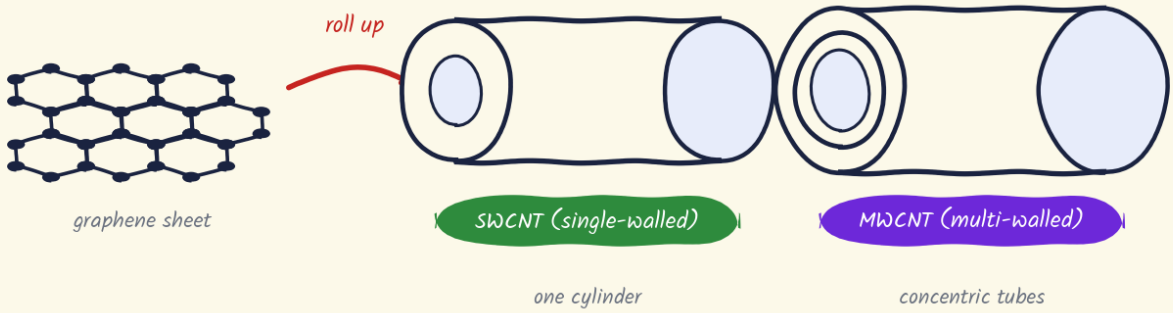
8. Carbon Nanotubes (CNTs)

Take a sheet of graphene and roll it into a cylinder — you get a carbon nanotube, one of the strongest materials ever made.

DEFINITION

Carbon nanotube (CNT): a rolled-up cylinder of graphene (sp^2 carbon). It may be **single-walled (SWCNT)** — one tube — or **multi-walled (MWCNT)** — several concentric tubes.

Carbon nanotube: a rolled-up graphene sheet



~100× stronger than steel, about one-sixth the weight · great conductor

Fig 7 — A graphene sheet rolls into a tube: SWCNT (one wall) and MWCNT (several concentric walls).

Properties

- About **100× the tensile strength of steel at only one-sixth the weight**.
- Excellent **electrical and thermal conductors**.
- Very high stiffness combined with flexibility.

Applications

- **Composites** — strong, light materials for sports gear and aerospace.
- **Nano-electronics** — transistors and field-emission displays.
- **Sensors** and **hydrogen storage**.
- **Electrodes** for batteries and supercapacitors.

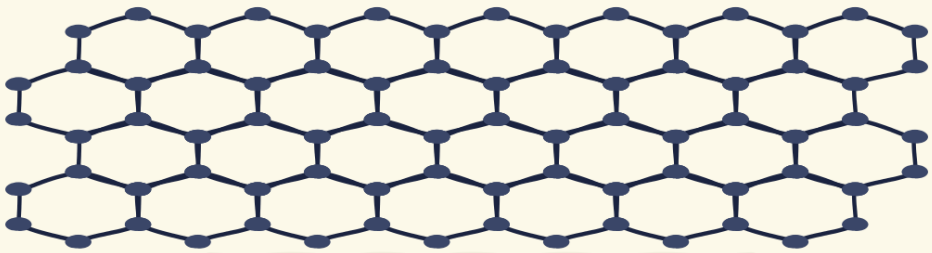
9. Graphene

Graphene is a single layer of carbon atoms — just one atom thick — yet it is the strongest material ever measured. Its discoverers won the 2010 Nobel Prize in Physics.

DEFINITION

Graphene: a single 2-D sheet of sp^2 carbon atoms arranged in a honeycomb (hexagonal) lattice — the basic building block of graphite, carbon nanotubes and fullerenes.

Graphene: a one-atom-thick honeycomb of sp^2 carbon



- each dot = C atom
- lines = C–C bonds
- 2-D sheet, 1 atom thick
- strongest known material

transparent · flexible · excellent heat & electrical conductor

Fig 8 — Graphene: a one-atom-thick honeycomb sheet of sp^2 carbon atoms.

Properties

- Strongest known material, yet light and flexible.
- Excellent electrical and thermal conductor.
- Transparent — lets light pass through.

Applications

- Fast transistors and flexible, transparent electronics.
- Sensors and battery and supercapacitor electrodes.

- Composites and desalination membranes.

> KEY POINT
> Graphene, CNTs and fullerenes are all made of the same sp^2 carbon — just shaped differently: a flat sheet, a rolled tube and a closed ball.

10. Applications & Importance

Together, these modern materials power nearly every advanced technology around us:

- Electronics & computing: semiconductor chips, transistors, LEDs and solar cells.
- Medicine: superconducting MRI magnets; fullerene drug delivery; CNT and graphene biosensors.
- Transport: Maglev trains, supercapacitor regenerative braking, and light CNT and graphene composites.
- Energy: supercapacitors and graphene electrodes for fast storage; photovoltaic cells; lossless power lines.
- Everyday materials: stronger, lighter sports and aerospace parts from nanocomposites.

QUICK RECAP

- Semiconductors: small band gap (Si 1.1 eV); conductivity rises with temperature and doping.
- n-type = group-15 doping (extra electrons, donor level); p-type = group-13 doping (holes, acceptor level).
- Superconductors: zero resistance below T_c, expel magnetic fields (Meissner effect); Cooper pairs (BCS).
- Supercapacitors: fast charge, very long life; types EDLC, pseudocapacitor, hybrid.
- Nanomaterials (1–100 nm): huge surface area + quantum effects; types 0-D, 1-D, 2-D, 3-D.
- Carbon trio: fullerene C_{60} (cage), CNT (rolled tube, very strong), graphene (one-atom sheet, strongest).

EXAM-ORIENTED QUESTIONS

Q1. What is a semiconductor? How does its conductivity change with temperature? [3 marks]

Ans. A semiconductor has conductivity between a conductor and an insulator, set by a small band gap (~0.5–3 eV). Unlike metals, its conductivity increases with temperature because heat excites more electrons across the gap into the conduction band.

Q2. Explain band theory for conductors, semiconductors and insulators. [5 marks]

Ans. Solids have a filled valence band and an empty conduction band separated by a band gap E_g. Conductors: bands overlap (no gap). Semiconductors: small gap (~1 eV). Insulators: large gap (>5 eV, e.g. diamond).

Q3. Distinguish between intrinsic and extrinsic semiconductors. [3 marks]

Ans. Intrinsic = pure Si or Ge with equal electrons and holes and low conductivity. Extrinsic = doped with a small impurity to raise conductivity, giving n-type or p-type.

Q4. Differentiate n-type and p-type semiconductors. [5 marks]

Ans. n-type: doped with group-15 (P, As); extra electrons are majority carriers; donor level lies just below the conduction band. p-type: doped with group-13 (B, Al); holes are majority carriers; acceptor level lies just above the valence band.

Q5. What is a superconductor and the critical temperature T_c? [3 marks]

Ans. A superconductor has exactly zero electrical resistance below a critical temperature T_c. It was discovered by Onnes (1911) in mercury at T_c = 4.2 K.

Q6. State the Meissner effect and give one application. [3 marks]

Ans. Below T_c a superconductor completely expels magnetic field lines (perfect diamagnetism), so a magnet levitates above it. It is the basis of Maglev trains and is used in MRI magnets.

Q7. Give the BCS idea and compare Type I and Type II superconductors. [5 marks]

Ans. BCS: below T_c electrons form Cooper pairs that move without resistance. Type I (soft, pure metals, very low T_c) expels the field fully; Type II (hard alloys and ceramics, higher T_c such as YBCO ≈ 92 K) is used in powerful magnets.

Q8. What is a supercapacitor? Name its three types. [3 marks]

Ans. An electrochemical device that stores charge mostly electrostatically on high-area electrodes — very fast charge-discharge and very long life, bridging a capacitor and a battery. Types: EDLC, pseudocapacitor and hybrid.

Q9. Define nanomaterials and give their classification by dimension. [5 marks]

Ans. Materials with at least one dimension in 1–100 nm, giving a huge surface area and quantum effects. Classified as 0-D (nanoparticles, quantum dots), 1-D (nanotubes, nanowires), 2-D (graphene, sheets) and 3-D (bulk nanostructured).

Q10. Describe the structure of fullerene C_{60} and give two uses. *[3 marks]*

Ans. C_{60} is a closed hollow cage of 60 sp^2 carbons (20 hexagons + 12 pentagons), shaped like a soccer ball. Uses: drug delivery and catalysts or organic solar cells.

Q11. What are carbon nanotubes? Compare SWCNT and MWCNT and give one property. *[3 marks]*

Ans. CNTs are rolled-up graphene cylinders. SWCNT has a single wall; MWCNT has several concentric walls. They are about 100× stronger than steel at one-sixth the weight and are excellent conductors.

Engineering Chemistry

UNIT – 3 : Electrochemistry and Applications

Cells, Nernst, titrations, sensors, batteries and fuel cells — the chemistry that powers your phone and your future, explained simply with clear diagrams.

Topics in this unit

1. Electrochemical cells & cell potential
2. Nernst equation & numericals
3. Potentiometry & conductometric titrations
4. Electrochemical sensors
5. Batteries: Zn-air & Li-ion
6. Fuel cells & PEMFC

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. What is Electrochemistry?

Every time you switch on a phone, start a car or flick a torch, you are using electrochemistry — the branch of chemistry that links chemical reactions and electricity. It studies how a chemical reaction can produce an electric current, and how electricity can drive a chemical change.

DEFINITION

Electrochemistry: the study of the relationship between chemical energy and electrical energy — how spontaneous redox reactions generate electricity, and how electricity can force non-spontaneous reactions to happen.

Redox — the heart of it all

Every electrochemical change is a redox (reduction–oxidation) reaction, in which electrons move from one species to another.

- Oxidation = loss of electrons — happens at the anode.
- Reduction = gain of electrons — happens at the cathode.
- Memory hook: OIL RIG → Oxidation Is Loss, Reduction Is Gain.

NOTE

Oxidation and reduction always occur together — the electrons lost by one species are exactly the electrons gained by another. You can never have one without the other.

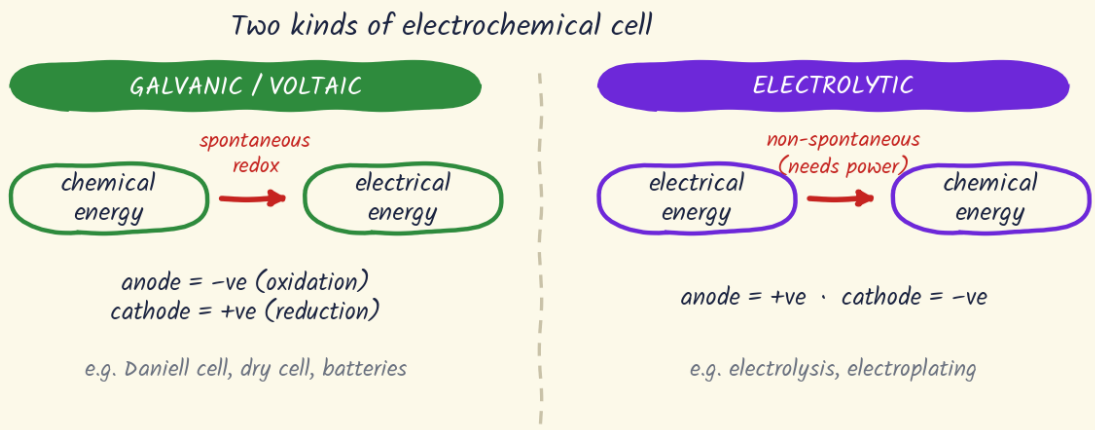


Fig 1 — A galvanic cell turns chemical energy into electricity (spontaneous); an electrolytic cell uses electricity to force a reaction.

2. Electrochemical Cells

An **electrochemical cell** is a device that either produces electricity from a chemical reaction, or uses electricity to carry one out. There are two families.

> DEFINITION
> **Galvanic (Voltaic) cell:** a cell that converts **chemical energy** into **electrical energy** using a **spontaneous** redox reaction. All batteries are galvanic cells.

How a galvanic cell works

- The two half-reactions are separated into two **half-cells**, so electrons are forced to travel through an external wire — that flow is the **current**.
- **Anode** (oxidation) is the **negative** terminal; **cathode** (reduction) is the **positive** terminal.
- A **salt bridge** (KCl or KNO_3) completes the circuit and keeps each solution electrically neutral.

The Daniell cell — a classic example

The Daniell cell uses a zinc electrode in $ZnSO_4$ and a copper electrode in $CuSO_4$. Zinc loses electrons more easily than copper, so zinc is the anode and copper is the cathode.

$$\text{Anode: } Zn \rightarrow Zn^{2+} + 2e^- \quad \text{Cathode: } Cu^{2+} + 2e^- \rightarrow Cu$$

Overall: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

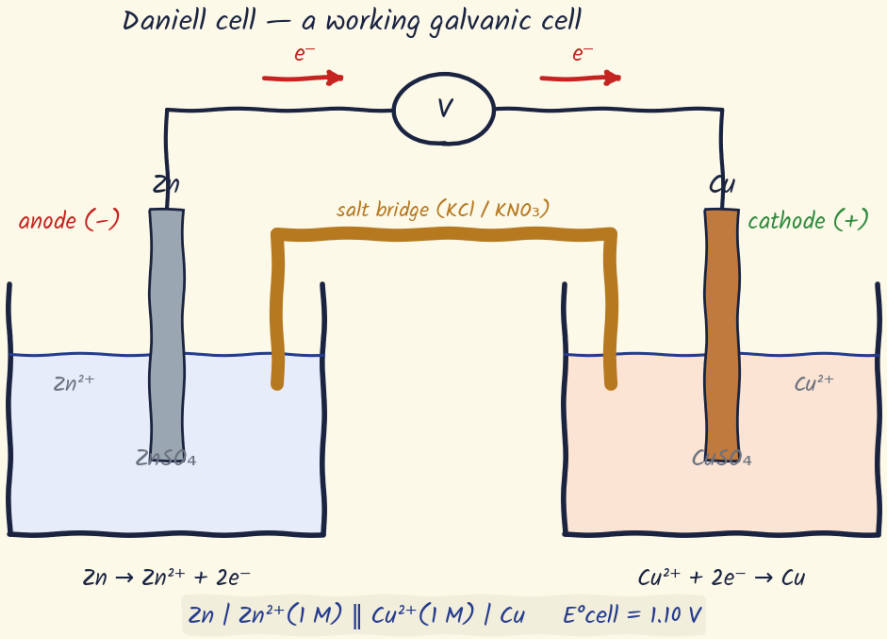


Fig 2 — The Daniell cell: electrons flow Zn → Cu through the wire while the salt bridge carries ions and keeps both beakers neutral.

> NOTE
> **Cell notation:** we write anode on the left, cathode on the right — $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$. A single bar | is a phase boundary (electrode–solution); the double bar ‖ is the salt bridge.

Electrolytic cell

DEFINITION

Electrolytic cell: a cell that uses electrical energy to drive a non-spontaneous chemical reaction (e.g. electrolysis, electroplating). Here the anode is positive and the cathode is negative — the opposite of a galvanic cell.

3. Electrode Potential & Cell Potential

Why do electrons flow from zinc to copper and not the other way? Because every electrode has a tendency to lose or gain electrons, measured as its **electrode potential**.

DEFINITION

Electrode potential: the tendency of an electrode to lose or gain electrons when dipped in its own ion solution. Measured against a reference, under standard conditions (1 M, 298 K, 1 atm) it is the **standard electrode potential (E°)**.

NOTE

Electrode potentials are measured against the **Standard Hydrogen Electrode (SHE)**, whose potential is fixed at $E^\circ = 0.00$ V by definition. Values are listed as standard **reduction** potentials — together they form the **electrochemical series**.

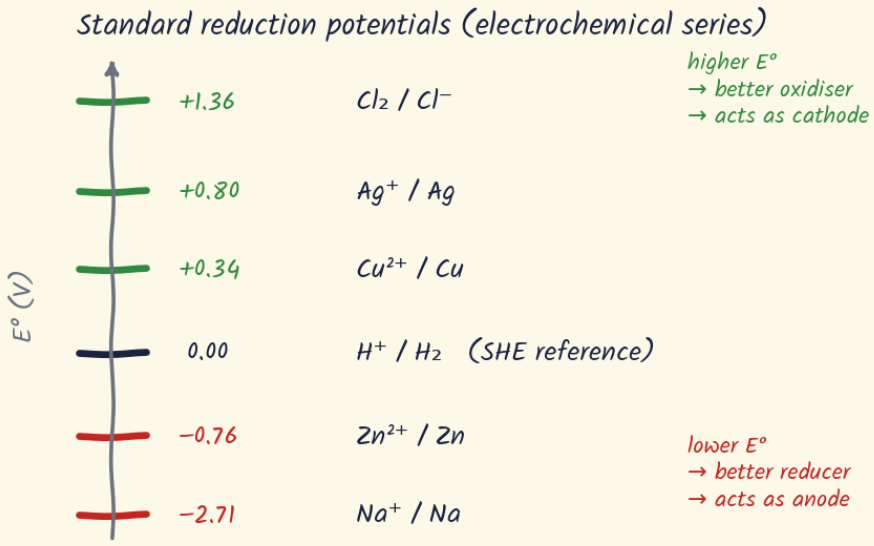


Fig 3 — The electrochemical series. The electrode with the higher E° acts as the cathode; the one with the lower E° acts as the anode.

Cell potential (EMF)

The cell potential or EMF is the voltage of the whole cell — the difference between the two electrode potentials. A positive E°cell means the reaction is spontaneous, so the cell can deliver current.

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

Both values are standard reduction potentials taken from the electrochemical series.

4. The Nernst Equation

Standard potentials assume every ion is at 1 M. In real cells the concentrations differ, and that changes the voltage. The Nernst equation tells us exactly how electrode and cell potentials depend on ion concentration and temperature.

For a single electrode

$$E = E^\circ + (0.0591/n) \log [M^{n+}] \text{ (at 298 K)}$$

n = number of electrons transferred; $[M^{n+}]$ = ion concentration

For the whole cell

$$E_{cell} = E^\circ_{cell} - (0.0591/n) \log Q \text{ (at 298 K)}$$

Q = reaction quotient; R = 8.314 J K⁻¹mol⁻¹, F = 96500 C, T = 298 K are built into 0.0591

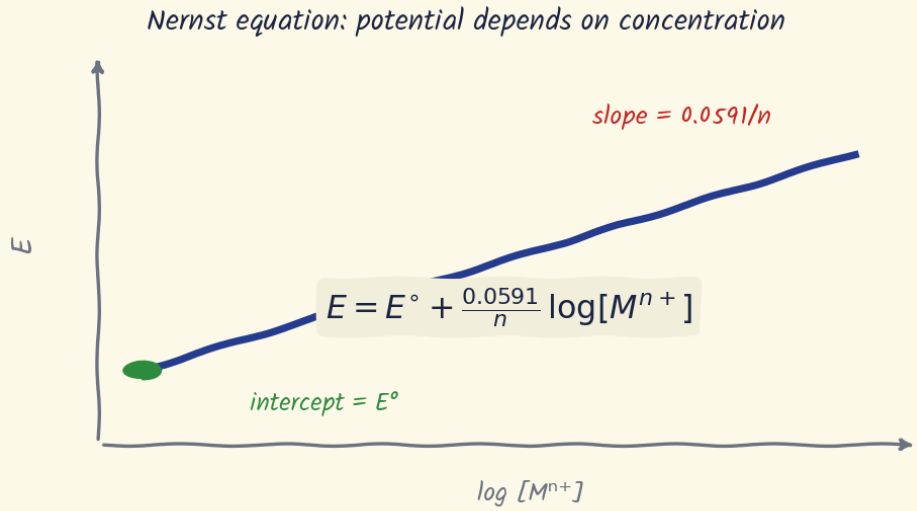


Fig 4 — Electrode potential varies linearly with log[ion]: the intercept is E° and the slope is 0.0591/n.

Worked examples

EXAMPLE 1

1. Standard EMF of the Daniell cell.

Given $E^\circ(Cu^{2+}/Cu) = +0.34$ V, $E^\circ(Zn^{2+}/Zn) = -0.76$ V.

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-0.76) = 1.10$ V.

EXAMPLE 2

2. EMF at non-standard concentration.

For $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, take $[Zn^{2+}] = 1.0$ M, $[Cu^{2+}] = 0.01$ M, $n = 2$, $T = 298$ K.

Step 1 — $Q = [Zn^{2+}] / [Cu^{2+}] = 1.0 / 0.01 = 100$.

Step 2 — $E_{cell} = 1.10 - (0.0591/2) \log(100) = 1.10 - (0.02955 \times 2)$.

Step 3 — $E_{cell} = 1.10 - 0.0591 = 1.04$ V. Lower $[Cu^{2+}]$ slightly lowers the EMF, exactly as Nernst predicts.

KEY POINT

When all ions are at 1 M, $\log Q = \log 1 = 0$, so $E_{cell} = E^{\circ}_{cell}$. The Nernst term only matters away from standard conditions.

5. Potentiometry & Potentiometric Titration

Instead of using a colour-change indicator, we can follow a titration by measuring voltage. This is potentiometry — and it is far more precise, especially for coloured or cloudy solutions.

DEFINITION

Potentiometric titration: a titration in which the EMF (or indicator-electrode potential) is measured after each addition of titrant. The equivalence point is found from a sharp jump in potential — no chemical indicator is needed.

- Plot E vs volume of titrant → an S-shaped (sigmoid) curve.
- Plot dE/dV vs volume → a sharp peak that pin-points the end point.
- Uses an indicator electrode (e.g. Pt) with a reference electrode (calomel).

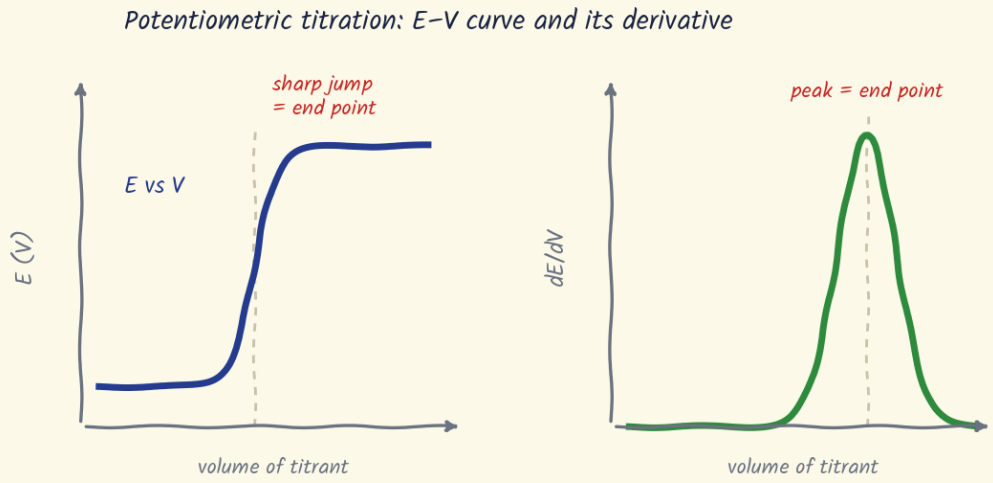


Fig 5 — The potential jumps sharply at the equivalence point; the derivative dE/dV peaks exactly there.

EXAMPLE

A redox titration of Fe^{2+} with $K_2Cr_2O_7$ (or Ce^{4+}) is followed with a Pt indicator electrode and a calomel reference. At the end point the electrode potential suddenly jumps as the last Fe^{2+} is used up.

NOTE

Advantages: no indicator needed, works for coloured/turbid solutions, very accurate, and can be automated.

6. Conductivity & Conductometric Titration

Ions carry current, so a solution's ability to conduct electricity depends on how many ions it has and how fast they move. Measuring this conductance is another neat way to find an end point.

DEFINITION

Specific conductance (κ): the conductance of a solution held between two electrodes 1 cm apart with 1 cm^2 area. Units: $S\ cm^{-1}$.

DEFINITION

Molar conductivity (Λ_m): the conducting power of all the ions from 1 mole of electrolyte.

$$\Lambda_m = \kappa \times 1000 / C \text{ cell constant} = l / A$$

C = concentration (mol L^{-1}); l = distance between electrodes; A = electrode area

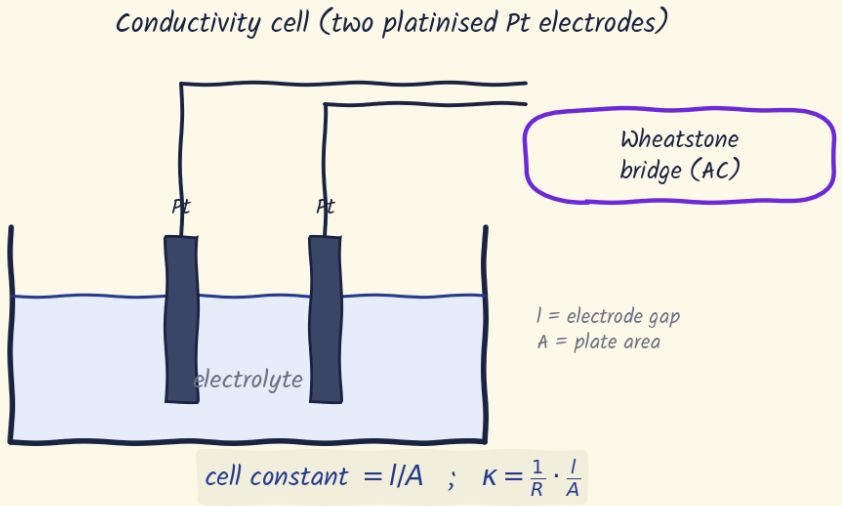


Fig 6 — A conductivity cell: two platinised Pt electrodes; resistance R is measured on an AC Wheatstone bridge.

Conductometric titration (strong acid vs strong base)

As we add NaOH to HCl, the conductance changes in a very telling way.

The fast, highly-conducting H^+ ions are neutralised and replaced by slower Na^+ ions.

- **Before the end point:** conductance falls — fast H^+ replaced by slower Na^+.
- **At the end point:** conductance reaches a minimum (only salt + water left).
- **After the end point:** conductance rises again — excess fast OH^- ions are now added.

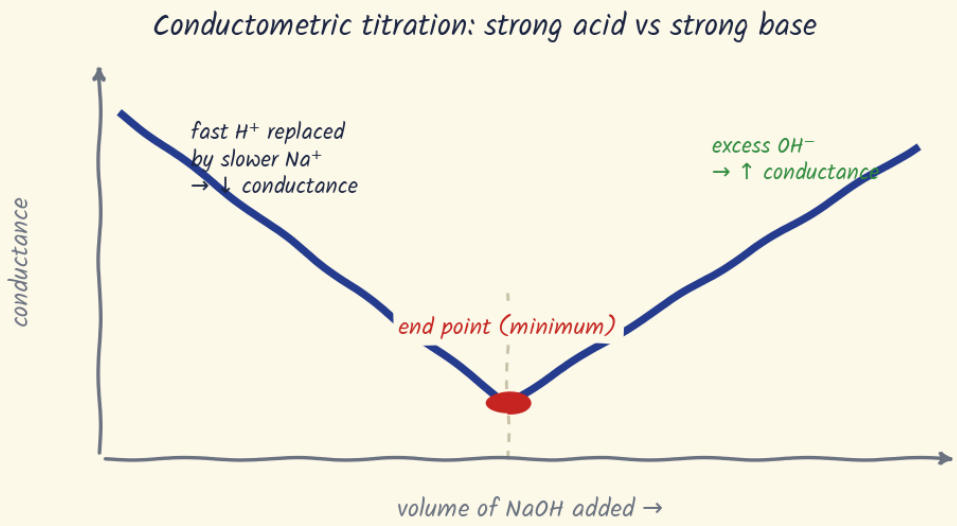


Fig 7 — The V-shaped curve: the end point is the minimum where the two straight lines meet.

KEY POINT

The end point is read from the break (minimum) of the V-shaped graph, not from any indicator — so even weakly coloured or dilute solutions can be titrated accurately.

7. Electrochemical Sensors

An electrochemical sensor turns the concentration of a chemical into an electrical signal we can read. They are cheap, fast and portable — think of the glucose meter used by diabetics.

DEFINITION

Electrochemical sensor: a device that converts a chemical quantity (concentration of an analyte) into a measurable electrical signal (voltage or current). The two main types are potentiometric and amperometric.

Potentiometric sensors

These measure a potential at (almost) zero current. The voltage is related to analyte concentration by the Nernst equation.

- pH glass electrode — potential depends on H^+ concentration.
- Ion-selective electrodes (ISE) — e.g. fluoride (LaF_3 crystal) and Ca^{2+} electrodes.

Amperometric sensors

These apply a fixed potential and measure the resulting current, which is directly proportional to analyte concentration.

- Clark oxygen electrode — measures dissolved O_2.
- Glucose biosensor — uses the enzyme glucose oxidase to measure blood glucose.

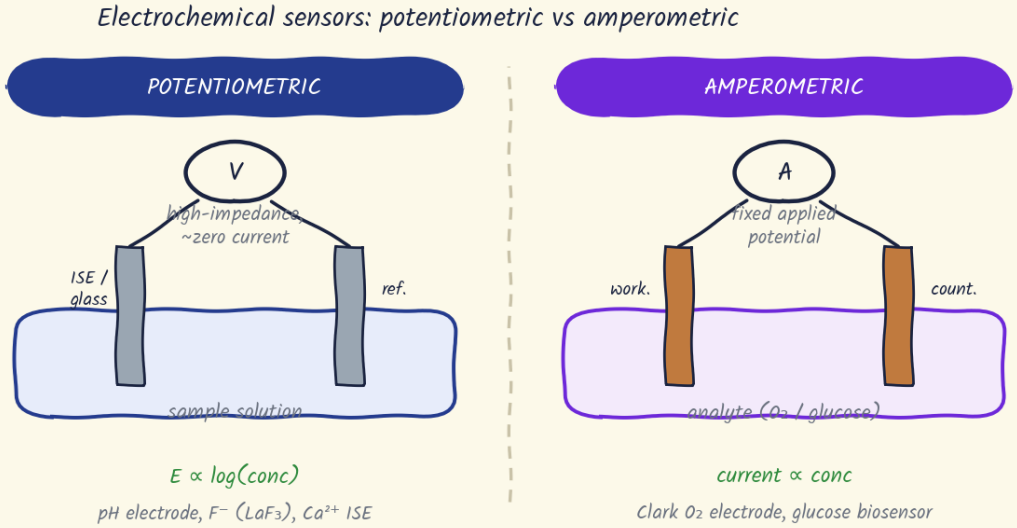


Fig 8 — Potentiometric sensors read a voltage at zero current ($E \propto \log c$); amperometric sensors read a current at a fixed voltage ($i \propto c$).

Feature	Potentiometric	Amperometric
Measured	potential (voltage)	current
Condition	~zero current	fixed applied potential
Relation	$E \propto \log(\text{conc})$ (Nernst)	current ∝ conc

Feature	Potentiometric	Amperometric
Examples	pH, F^- (LaF_3), Ca^{2+} ISE	Clark O_2, glucose sensor

8. Batteries: Zinc–Air & Li-ion

A battery is just one or more galvanic cells packaged to store and deliver electricity. Primary batteries are used once; secondary batteries can be recharged.

DEFINITION

Primary vs secondary: Primary cells give current until the reactants run out and are then thrown away (non-rechargeable). Secondary cells use a reversible reaction, so they can be recharged many times.

Zinc–air battery (primary)

A high-energy primary cell: a zinc anode, oxygen from the air as the cathode reactant on a porous carbon electrode, and a KOH electrolyte.

$$\text{Anode: } Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$$

Cathode: $\tfrac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$; Overall: $Zn + \tfrac{1}{2}O_2 \rightarrow ZnO$

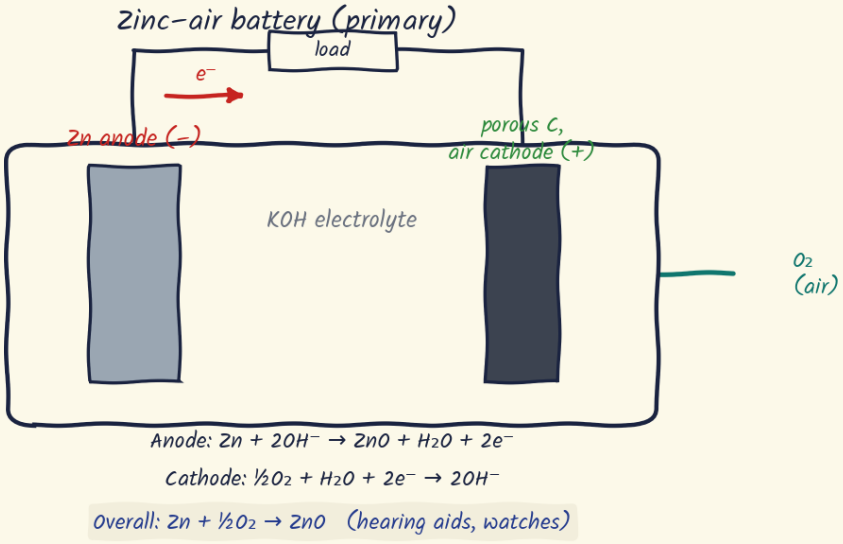


Fig 9 — Zinc–air cell: Zn is oxidised at the anode while oxygen from the air is reduced at the porous carbon cathode.

- **Very high energy density** because oxygen comes free from the air.
- **Uses:** hearing aids and watches.

Lithium-ion battery (secondary)

The rechargeable battery in your phone and laptop. The anode is **graphite** (LiC_6), the cathode is $LiCoO_2$, and the electrolyte is a lithium salt ($LiPF_6$) in an organic solvent with a separator.

> **NOTE**
> It is called a **"rocking-chair"** cell because Li^+ ions simply shuttle back and forth between the two electrodes on charge and discharge.

$$\text{Discharge anode: } LiC_6 \rightarrow Li^+ + e^- + 6C$$

Discharge cathode: $Li^+ + CoO_2 + e^- \rightarrow LiCoO_2$ (charging reverses both)

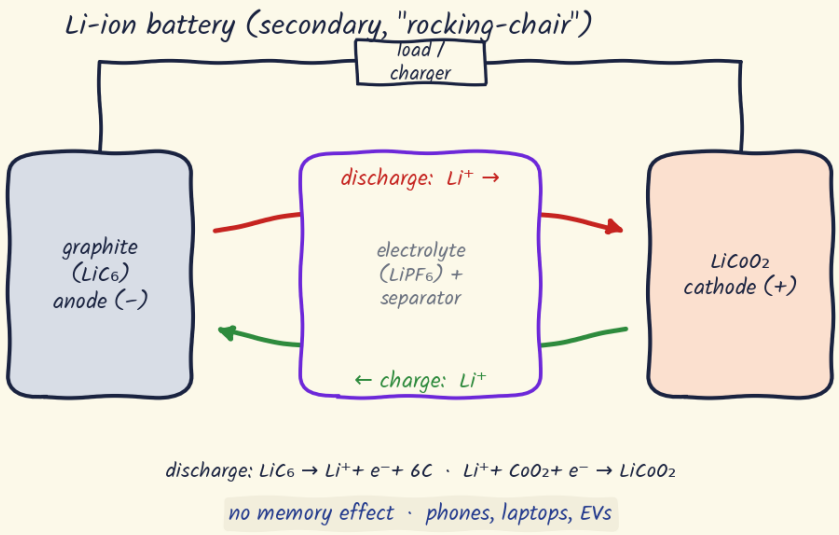


Fig 10 — Li-ion cell: Li^+ moves anode → cathode on discharge and back again on charging.

- **Uses:** phones, laptops and electric vehicles (EVs); **no memory effect**.

9. Fuel Cells

A **fuel cell** is like a battery that never runs down — as long as you keep feeding it fuel and oxygen, it keeps making electricity, cleanly and efficiently.

> DEFINITION
> **Fuel cell:** a device that converts the **chemical energy of a fuel directly and continuously into electricity** as long as fuel and an oxidant are supplied. It gives **high efficiency** and **low pollution**.

H_2–O_2 (alkaline) fuel cell

Hydrogen is the fuel and oxygen the oxidant, with a **KOH** electrolyte and **porous carbon** electrodes containing a **Pt / Ni** catalyst.

$$\text{Anode: } 2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$$

Cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; Overall: $2H_2 + O_2 \rightarrow 2H_2O$

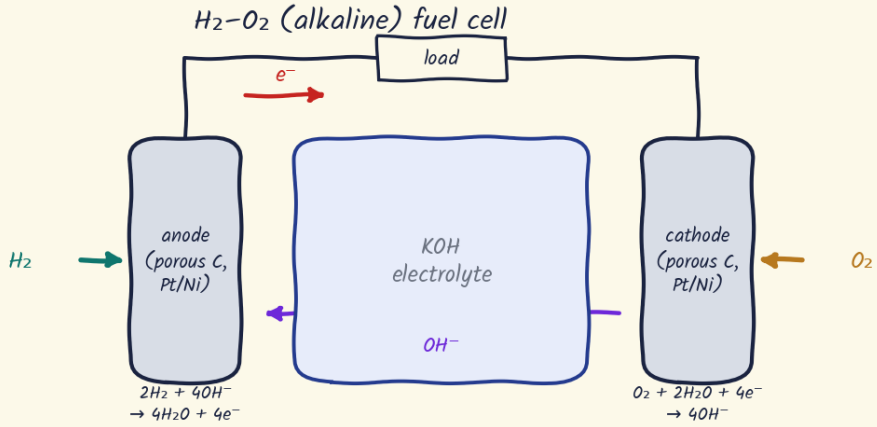


Fig 11 — H_2–O_2 alkaline fuel cell: the only product is water. Famously used in the Apollo spacecraft.

PEM fuel cell (PEMFC)

The Proton Exchange Membrane fuel cell replaces the liquid electrolyte with a solid polymer membrane (Nafion) that conducts H^+ ions. It runs cool (~60–80 °C) with a Pt catalyst.

$$\text{Anode: } H_2 \rightarrow 2H^+ + 2e^-$$

Cathode: $\tfrac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$ (H^+ crosses the membrane)

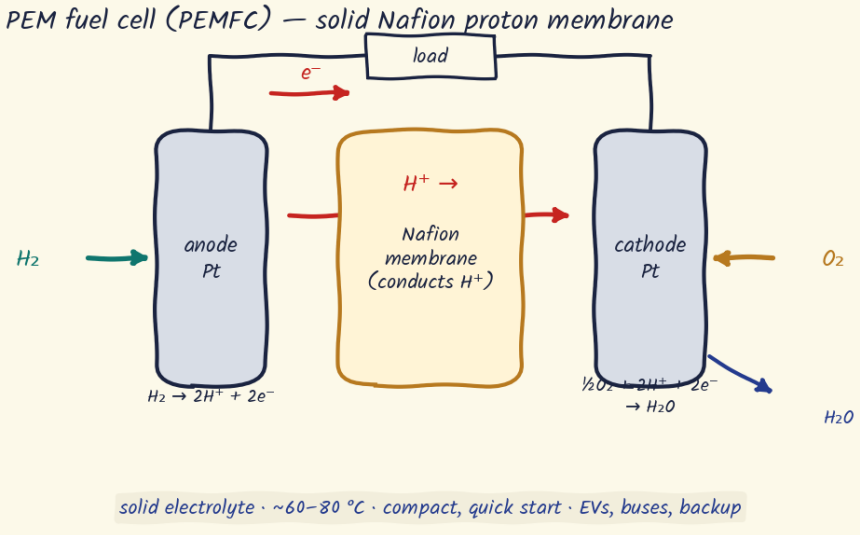


Fig 12 — PEMFC: protons cross the solid Nafion membrane while electrons drive the external load.

- **Advantages:** solid electrolyte (no leaks), compact and light, quick start-up.
- **Uses:** electric vehicles and buses, portable and backup power.

10. Applications & Importance

Electrochemistry quietly powers modern life:

- **Energy storage:** dry cells, Zn–air, lead-acid and Li-ion batteries for phones, laptops and EVs.
- **Clean power:** H_2–O_2 and PEM fuel cells for spacecraft, vehicles and backup supplies.
- **Measurement & sensing:** potentiometric/conductometric titrations, pH meters, ion-selective electrodes, Clark O_2 and glucose biosensors.
- **Industry:** electroplating, electro-refining of metals and corrosion protection.

QUICK RECAP

- Electrochemistry links chemical and electrical energy through redox reactions (oxidation at the anode, reduction at the cathode).
- A galvanic cell (e.g. Daniell, $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$) makes electricity spontaneously; an electrolytic cell uses electricity.
- $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$; a positive value means a spontaneous cell (Daniell = +1.10 V).
- Nernst: $E_{cell} = E^\circ_{cell} - (0.0591/n) \log Q$ — potential depends on concentration.
- Potentiometric titration → sigmoid E–V with a sharp jump; conductometric (strong acid–base) → V-shaped curve, end point at the minimum.
- Sensors: potentiometric (pH, ISE; $E \propto \log c$) and amperometric (Clark O_2, glucose; $i \propto c$).
- Batteries: Zn–air (primary), Li-ion (secondary, rocking-chair). Fuel cells: H_2–O_2 alkaline and PEMFC (solid Nafion membrane).

EXAM-ORIENTED QUESTIONS

Q1. Differentiate between a galvanic cell and an electrolytic cell. [3 marks]

Ans. A galvanic cell converts chemical energy to electrical energy using a spontaneous redox reaction (anode −, cathode +). An electrolytic cell uses electrical energy to drive a non-spontaneous reaction (anode +, cathode −).

Q2. Describe the construction and working of the Daniell cell with its cell notation. [5 marks]

Ans. A Zn electrode in $ZnSO_4$ (anode) and a Cu electrode in $CuSO_4$ (cathode) joined by a salt bridge. $Zn \rightarrow Zn^{2+} + 2e^-$; $Cu^{2+} + 2e^- \rightarrow Cu$. Electrons flow Zn → Cu externally. Notation: $Zn \mid Zn^{2+} \parallel Cu^{2+} \mid Cu$.

Q3. State the function of a salt bridge. [2 marks]

Ans. It completes the internal circuit and maintains electrical neutrality in both half-cells by allowing ions (from KCl/KNO_3) to migrate.

Q4. Write the Nernst equation for a cell and explain each term. [3 marks]

Ans. $E_{cell} = E^\circ_{cell} - (0.0591/n) \log Q$ at 298 K, where E°_{cell} = standard EMF, n = electrons transferred, Q = reaction quotient. It gives the EMF at non-standard concentrations.

Q5. Calculate the standard EMF of the Daniell cell. [2 marks]

Ans. $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-0.76) = +1.10$ V.

Q6. How is the equivalence point found in a potentiometric titration? [5 marks]

Ans. The cell EMF is measured after each addition of titrant and plotted against volume, giving a sigmoid curve with a sharp jump at the equivalence point. A dE/dV vs V plot shows a peak exactly at the end point. No indicator is required.

Q7. Draw and explain the conductometric titration curve of a strong acid vs a strong base. [5 marks]

Ans. Conductance first falls as fast H^+ ions are replaced by slower Na^+, reaching a minimum at the end point; then it rises as excess fast OH^- is added — giving a V-shaped curve. The end point is the minimum.

Q8. Distinguish between potentiometric and amperometric sensors with examples. [3 marks]

Ans. Potentiometric sensors measure a potential at ~zero current, related to concentration by the Nernst equation (pH glass electrode, F^-/Ca^{2+} ISE). Amperometric sensors measure current at a fixed applied potential, proportional to concentration (Clark O_2 electrode, glucose biosensor).

Q9. Explain the working of a zinc-air battery with electrode reactions. [5 marks]

Ans. A primary cell: Zn anode, O_2 from air reduced on a porous carbon cathode, KOH electrolyte. Anode: $Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$; cathode: $\frac{1}{2}O_2 + H_2O + 2e^- \rightarrow 2OH^-$; overall $Zn + \frac{1}{2}O_2 \rightarrow ZnO$. Used in hearing aids and watches.

Q10. Why is a Li-ion battery called a "rocking-chair" cell? Give its electrodes. [3 marks]

Ans. Because Li^+ ions shuttle back and forth between the electrodes on charge and discharge. Anode = graphite (LiC_6), cathode = $LiCoO_2$, electrolyte = $LiPF_6$ in organic solvent. Rechargeable, no memory effect.

Q11. What is a fuel cell? Give the reactions of the H_2–O_2 (alkaline) fuel cell. [5 marks]

Ans. A cell that converts the chemical energy of a fuel directly and continuously to electricity while fuel + oxidant are supplied. Anode: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$; cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; overall $2H_2 + O_2 \rightarrow 2H_2O$. Used in Apollo spacecraft.

Q12. Write a short note on the PEM fuel cell (PEMFC). [3 marks]

Ans. It uses a solid proton-exchange membrane (Nafion) that conducts H^+, a Pt catalyst and runs at ~60–80 °C. Anode: $H_2 \rightarrow 2H^+ + 2e^-$; cathode: $\tfrac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$. Compact, solid electrolyte, quick start; used in EVs and backup power.

Engineering Chemistry

UNIT – 4 : Polymer Chemistry

From monomers to materials — how everyday plastics, rubbers and smart polymers are built, with clear repeat-unit sketches.

Topics in this unit

1. Polymers, functionality & polymerization
2. Chain, step & coordination polymerization
3. Plastics: PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres
4. Elastomers: Buna-S & Buna-N
5. Conducting polymers
6. Biodegradable polymers (PGA, PLA)

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. What is a Polymer?

Look around you — bottles, pipes, clothes, tyres, phone cases. Almost all of them are made of polymers. A polymer is a very large molecule built by joining hundreds or thousands of small, repeating units together like beads on a string.

DEFINITION

Polymer: a large molecule (a macromolecule) made of many small repeating units called monomers, joined together by strong covalent bonds.

DEFINITION

Polymerization: the chemical process that joins many monomer molecules into a polymer chain.

A few words you will meet again and again:

- **Monomer** — the small starting molecule (e.g. ethene, $CH_2=CH_2$).
- **Repeat unit** — the part that repeats over and over inside the chain.
- **Degree of polymerization (n)** — the number of repeat units in one chain. A big n means a long, heavy chain.

$$n\ CH_2=CH_2 \rightarrow -(CH_2-CH_2)_n-$$

n molecules of ethene join to give polyethylene (polythene)

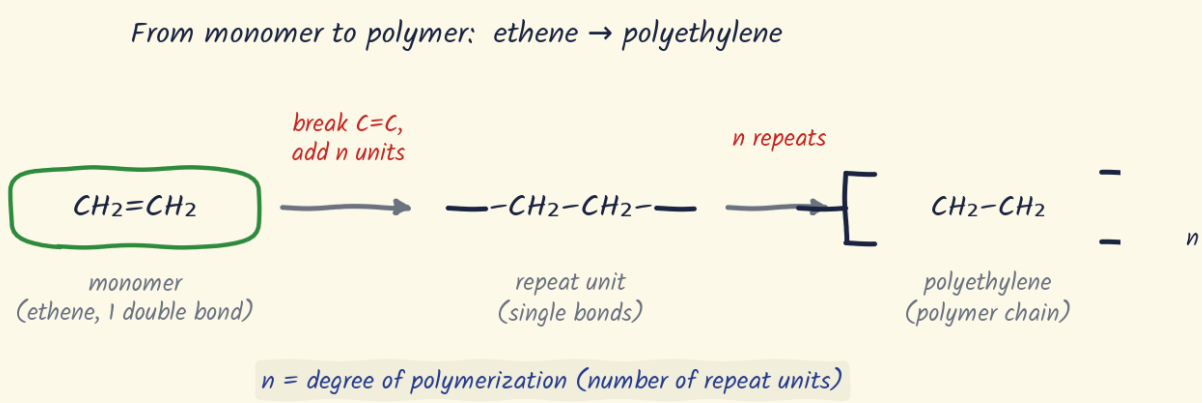


Fig 1 — One monomer opens its double bond, becomes a repeat unit, and n of them link into a long polymer chain.

EXAMPLE

Polyethylene: n molecules of ethene ($CH_2=CH_2$) join end-to-end to form $-(CH_2-CH_2)_n-$ — the plastic used in carry-bags and bottles.

NOTE

Natural polymers also exist — **starch**, **cellulose**, **proteins** and **DNA** are all polymers made by nature. In this unit we focus mainly on synthetic (man-made) polymers.

2. Functionality of a Monomer

Why do some polymers form soft, flexible threads while others set into hard, rigid solids? The answer is **functionality** — how many bonding arms each monomer has.

DEFINITION

Functionality: the number of **reactive sites** (bonding points) on a monomer that can join to other monomers.

The functionality decides the shape of the final polymer:

- Functionality = 2 → each monomer links to just two neighbours → a straight linear chain. Example: ethylene, adipic acid.
- Functionality ≥ 3 → monomers link in several directions → a branched or cross-linked network. Example: glycerol; phenol + formaldehyde.

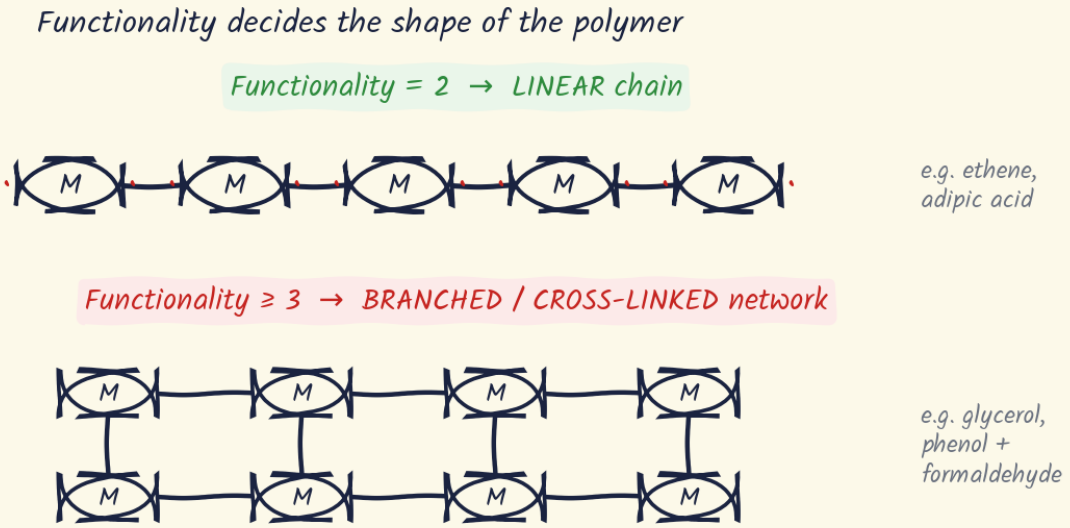


Fig 2 — Bifunctional monomers give a linear chain; tri- (or higher) functional monomers build a cross-linked network.

EXAMPLE

Ethene has functionality 2 (one double bond opens into two bonds) → linear polythene. Phenol + formaldehyde gives functionality ≥ 3 → the rigid 3-D network of Bakelite.

KEY POINT

Simple rule: 2 arms → thread, 3 or more arms → net. Linear polymers are usually soft and remouldable; cross-linked ones are hard and permanent.

3. Chain-Growth (Addition) Polymerization

There are three main ways to build a polymer. The first is **chain-growth**, also called **addition** polymerization.

DEFINITION

Chain-growth polymerization: unsaturated monomers (with a C=C double bond) add on rapidly, one after another, through a reactive intermediate. **No small molecule is lost**, and a high molecular weight is reached very quickly.

- Needs monomers with a **double bond** (C=C), e.g. ethene, vinyl chloride.
- The chain grows **very fast** through a reactive centre (a free radical).
- **No by-product** is formed — every atom of the monomer ends up in the polymer.

Free-radical mechanism — three steps

- **Initiation** — an initiator (e.g. **benzoyl peroxide**) breaks into free radicals (R•) that attack a monomer.
- **Propagation** — the radical adds monomer after monomer, growing the chain rapidly.
- **Termination** — two growing chains join (**combination**) or swap an atom (**disproportionation**), and growth stops.

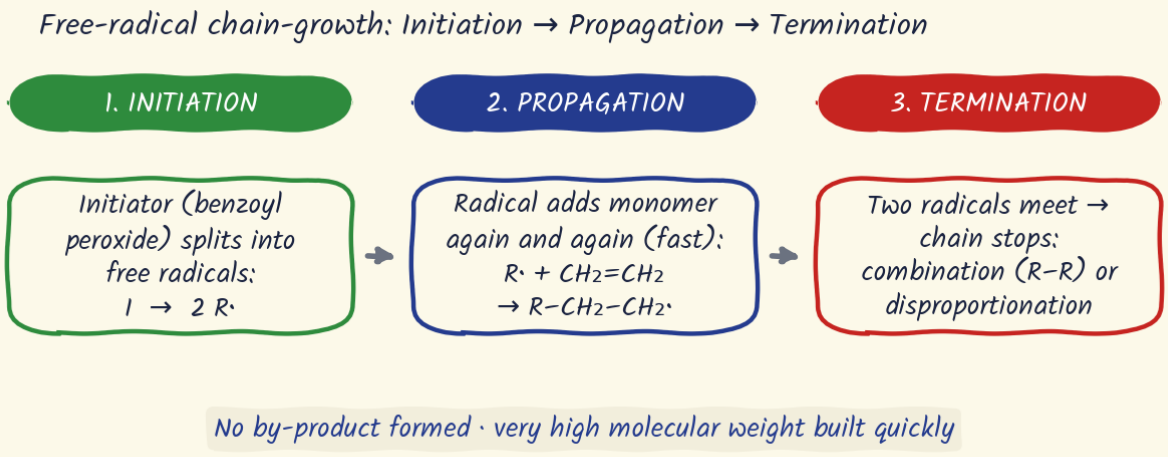


Fig 3 — The three steps of free-radical chain-growth polymerization.

EXAMPLE

Polyethylene is made this way: ethene ($CH_2=CH_2$) molecules add on one by one to give long $-(CH_2-CH_2)_n-$ chains.

4. Step-Growth (Condensation) Polymerization

The second route is step-growth, usually called condensation polymerization. Here growth is slow and steady, and a small molecule is thrown out at each step.

DEFINITION

Step-growth polymerization: bi-functional monomers react step by step; at each joint a small molecule (usually H_2O or HCl) is eliminated. Molecular weight builds up slowly.

- Monomers carry two reactive groups (e.g. –COOH, –OH, $-NH_2$).
- A small molecule is released at every linkage (condensation).
- Any two units can join (monomer+monomer, dimer+trimer…), so the chain grows gradually.

EXAMPLE

Nylon-6,6 (adipic acid + hexamethylenediamine), PET polyester and Bakelite (phenol + formaldehyde) are all made by condensation.

Chain-growth vs step-growth

Feature	Chain-growth (addition)	Step-growth (condensation)
Monomer	has a C=C double bond	bi-functional (–OH, –COOH, $–NH_2$)
Growth	very fast, adds one monomer at a time	slow, any two units combine
By-product	none	small molecule (H_2O / HCl)
Mol. weight	high almost immediately	builds up gradually
Examples	polythene, PVC, Teflon	Nylon-6,6, PET, Bakelite

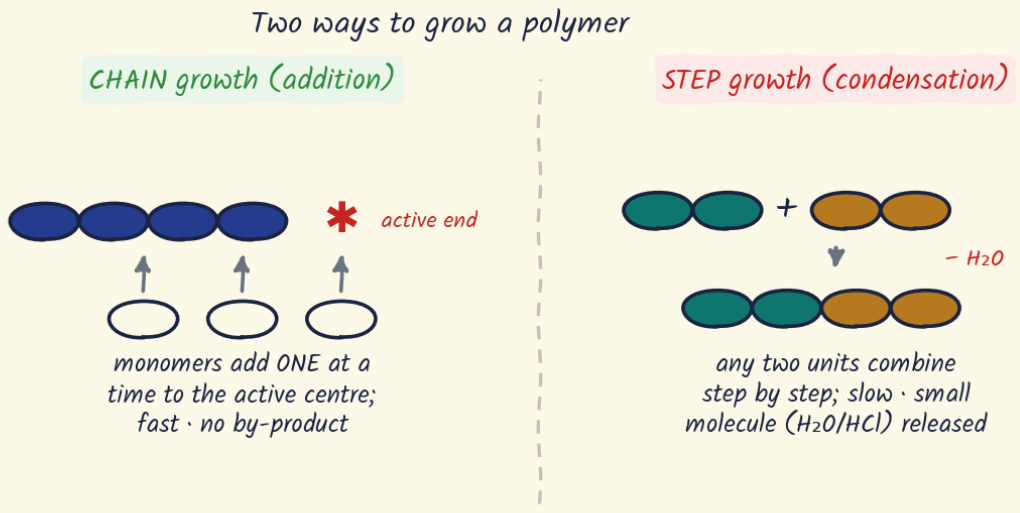


Fig 4 — Chain-growth adds single monomers to one active end; step-growth joins any two units and releases a small molecule.

5. Coordination Polymerization (Ziegler–Natta)

The third route uses a special catalyst to build very neat, well-ordered chains. It gave us modern high-quality plastics and won a Nobel Prize.

DEFINITION

Coordination polymerization: polymerization carried out with a **Ziegler–Natta catalyst** — $TiCl_4$ + $Al(C_2H_5)_3$ (triethylaluminium). It gives **stereoregular** (isotactic), linear, high-density polymers, and works at **low pressure**.

- Monomers insert one by one at the metal centre, so each unit lines up the same way.
- The product is **stereoregular (isotactic)** — side groups all on the same side.
- Chains are **linear and high-density**, giving stronger, denser plastics.

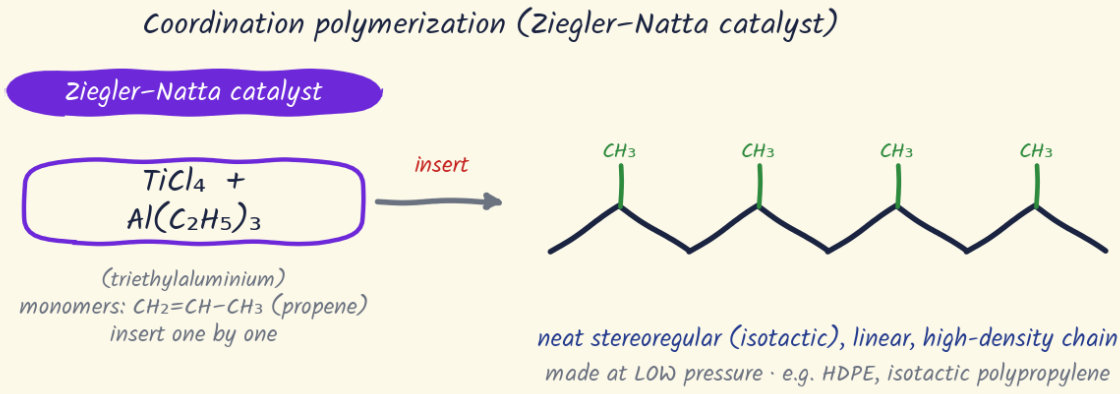


Fig 5 — A Ziegler–Natta catalyst lines monomers up into a neat, stereoregular, linear chain.

EXAMPLE

HDPE (high-density polyethylene) and **isotactic polypropylene** are made by coordination polymerization.

KEY POINT

Big advantage: the catalyst gives control over structure and stereochemistry, so the plastic is denser, more crystalline and much stronger.

6. Plastics: Thermoplastic vs Thermosetting

Plastics are polymers that can be moulded into shapes. They fall into two big families based on how they behave when heated.

DEFINITION

Thermoplastics: soften on heating and can be remoulded again and again. They have linear or branched chains held by weak forces. e.g. PVC, Teflon, polythene, nylon.

DEFINITION

Thermosetting plastics: set permanently hard on heating (a 3-D cross-linked network) and cannot be remoulded. e.g. Bakelite, epoxy.

Property	Thermoplastic	Thermosetting
Structure	linear/branched chains, weak forces	3-D cross-linked network
On heating	softens → can remould	sets hard → cannot remould
Reusable?	yes, recyclable	no
Examples	PVC, Teflon, polythene, nylon	Bakelite, epoxy resin

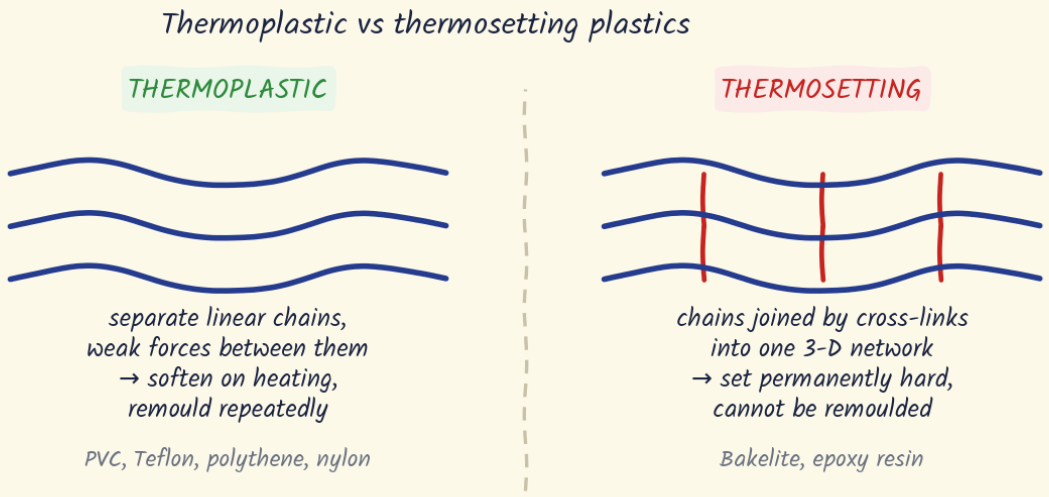


Fig 6 — Thermoplastics are separate chains that slide when heated; thermosets are one locked cross-linked network.

KEY POINT

Memory hook: thermo-PLASTIC stays plastic (re-softens); thermo-SET is set for good.

7. Important Plastics

Let us meet five everyday plastics, their monomers and where they are used.

PVC — Polyvinyl Chloride

Made by addition polymerization of vinyl chloride ($CH_2=CHCl$). It can be made rigid, or soft and flexible by adding a plasticizer.

- Uses: pipes, cable insulation, raincoats, flooring, bottles.

Teflon (PTFE)

Addition polymer of tetrafluoroethylene ($CF_2=CF_2$). The strong C–F bonds make it chemically inert, heat-resistant, non-stick and a good insulator

with very low friction.

- Uses: non-stick cookware coatings, gaskets and seals, electrical insulation.

Bakelite — Phenol-Formaldehyde

A thermosetting plastic made by condensation of phenol + formaldehyde into a 3-D cross-linked network. It is hard and resists both heat and electricity.

- Uses: electrical switches, plugs, handles, fittings.

Nylon-6,6 — a Polyamide

Made by condensation of adipic acid (6 carbons) + hexamethylenediamine (6 carbons) — hence the name 6,6. The chains are joined by amide linkages (–CO–NH–), so nylon is a polyamide. It is strong, elastic and abrasion-resistant.

- Uses: fibres, ropes, textiles, gears, bearings, carpets.

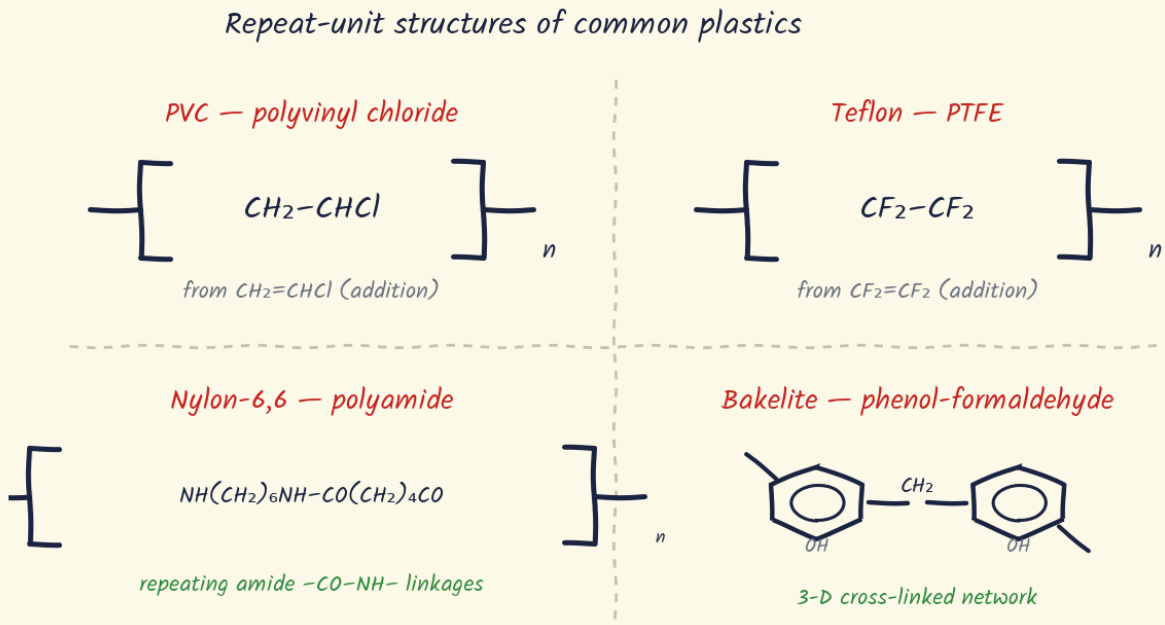


Fig 7 — Repeat-unit sketches: PVC, Teflon, Nylon-6,6 (note the –CO–NH– amide links) and the Bakelite network.

Carbon Fibres

Fibres of at least **90% carbon** made by the controlled **pyrolysis** (heating) of **PAN — polyacrylonitrile**. They have a very high strength-to-weight ratio, high stiffness and resist heat.

- **Uses:** aerospace parts, sports goods, automobiles, construction.

> EXAMPLE
> Carbon fibre is **lighter than aluminium but stiffer than steel** — which is why aircraft, racing cars and premium bicycles use it.

8. Elastomers: Buna-S and Buna-N

Elastomers are rubber-like polymers that stretch a lot and then snap back to shape. Two important synthetic rubbers are Buna-S and Buna-N — both **copolymers** (made from two different monomers) of

butadiene.

Buna-S (SBR)

A copolymer of butadiene + styrene, made by free-radical emulsion polymerization. It is tough and abrasion-resistant.

- Uses: automobile tyres, shoe soles, cable insulation, conveyor belts.

Buna-N (NBR)

A copolymer of butadiene + acrylonitrile. The –C≡N groups make it oil-, solvent- and heat-resistant.

- Uses: fuel hoses, oil seals, gaskets, protective gloves.

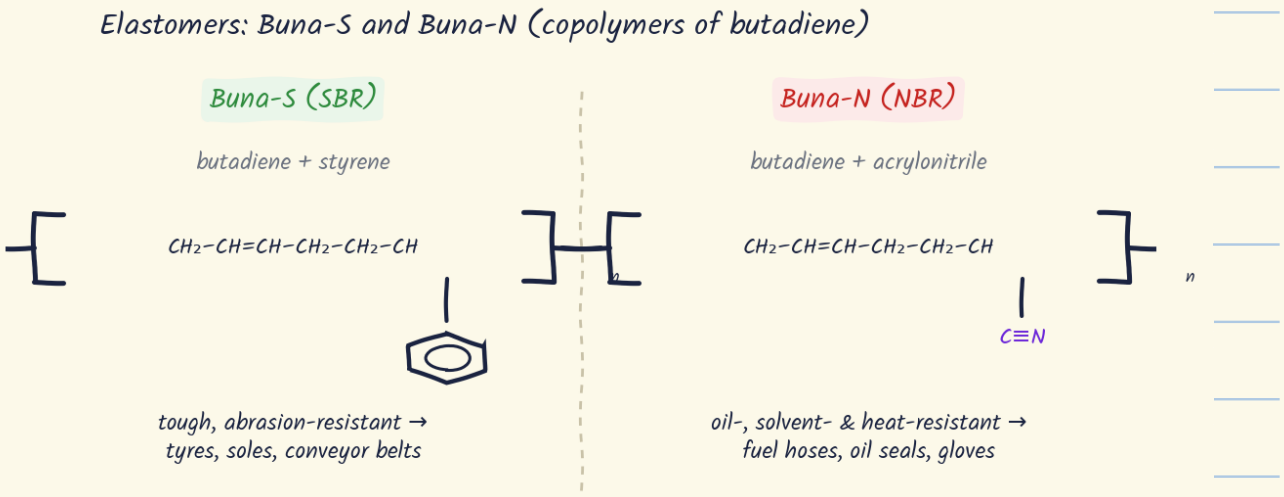


Fig 8 — Repeat-unit sketches of Buna-S (butadiene + styrene) and Buna-N (butadiene + acrylonitrile).

> EXAMPLE
>
> The name Buna comes from Butadiene + Na (sodium, the old catalyst). Buna-S has Styrene; Buna-N has the Nitrile group.

9. Conducting Polymers

Most plastics are insulators — but some special polymers can actually conduct electricity almost like a metal. These are the conducting polymers.

DEFINITION

Conducting polymer: a polymer that conducts electricity because of a conjugated backbone (alternating single and double bonds → delocalized π electrons) plus doping.

- Polyacetylene $(-CH=CH-)_n$ — doping with I_2 (oxidation, p-type) or Na (reduction, n-type) makes it highly conducting.
- Polyaniline — its conductivity is controlled by its oxidation state and protonation (acid doping).

How does it conduct?

The conjugated backbone gives a delocalized π pathway along the chain. Doping then removes or adds electrons, creating mobile charge carriers (polarons) that drift along the chain under an electric field → current flows.

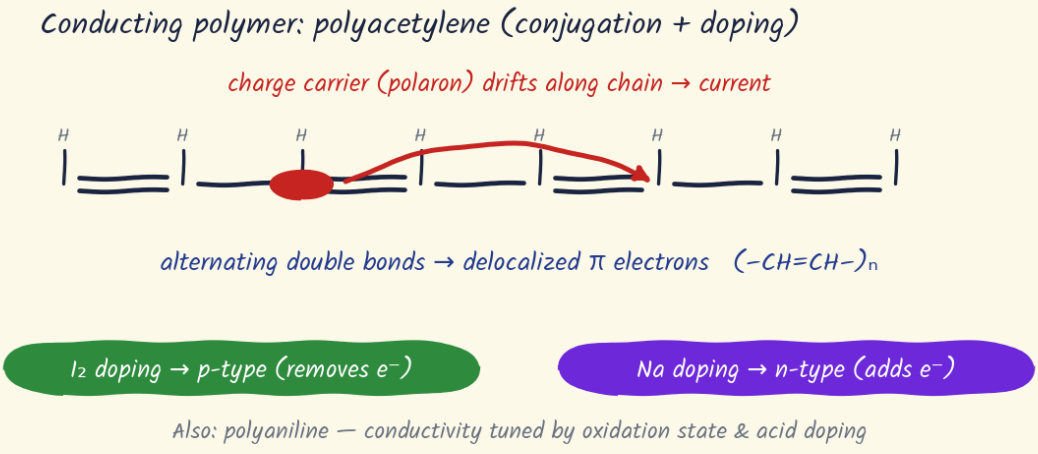


Fig 9 — A conjugated polyacetylene chain; doping creates a mobile charge carrier that drifts along the π system.

Applications:

- Rechargeable batteries and electrodes.
- Sensors and antistatic coatings.
- LEDs / OLEDs and electrochromic displays.
- EMI (electromagnetic interference) shielding.

KEY POINT

Two things are needed together: a conjugated backbone (the highway) and doping (the moving charges). Remove either and conduction stops.

10. Biodegradable Polymers

Ordinary plastics last for centuries and choke the environment. Biodegradable polymers are designed to break down naturally — a greener choice.

DEFINITION

Biodegradable polymer: a polymer that micro-organisms can break down into CO_2, H_2O and biomass, so it does not pile up as waste.

- PGA (polyglycolic acid) — made from glycolic acid by condensation. Used for dissolvable surgical sutures and tissue scaffolds.
- PLA (polylactic acid) — made from lactic acid (from corn starch / sugar fermentation). Used for packaging, cups, cutlery, 3-D printing filament, sutures, drug delivery and implants.

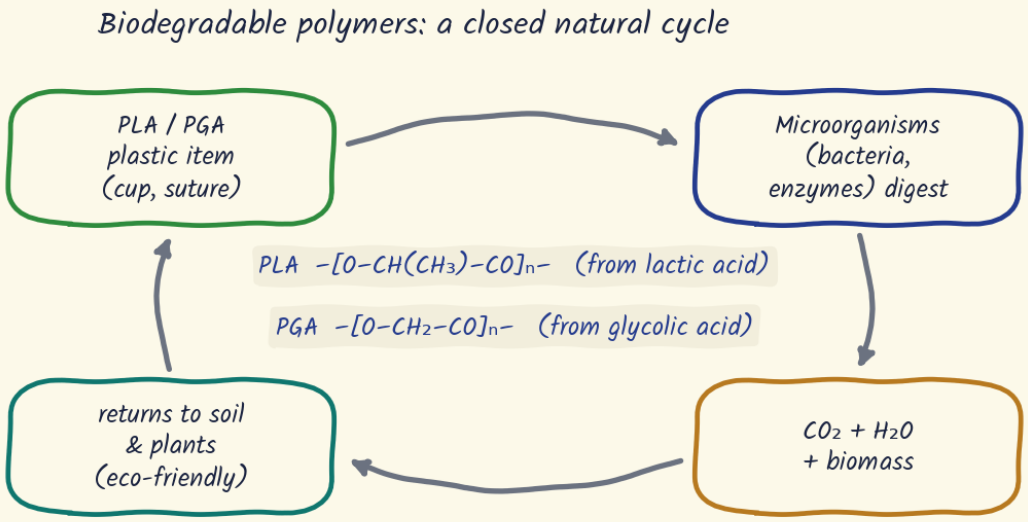


Fig 10 — A biodegradable polymer item is digested by micro-organisms into CO_2, water and biomass, returning to nature.

EXAMPLE

A PLA cup thrown into a compost heap is eaten by microbes and turns into CO_2, water and biomass — unlike a normal plastic cup that survives for hundreds of years.

II. Applications & Importance

Polymers are everywhere in modern engineering and daily life:

- **Packaging & daily use:** polythene bags, PVC bottles, PET water bottles.
- **Engineering parts:** nylon gears and bearings, HDPE pipes, carbon-fibre aerospace and automobile components.
- **Electrical insulation:** Bakelite switches, PVC cable sheathing, Teflon insulation.
- **Tyres & sealing:** Buna-S tyres and belts; Buna-N oil seals and fuel hoses.

- **Electronics:** conducting polymers in batteries, OLED displays, sensors and antistatic / EMI-shielding coatings.
- **Medicine & environment:** biodegradable PGA/PLA sutures and implants, eco-friendly compostable packaging.

QUICK RECAP

- A **polymer** is a macromolecule of many repeating **monomer** units; n = degree of polymerization.
- **Functionality** 2 → linear chain; **functionality** ≥ 3 → cross-linked network.
- **Chain-growth** (addition) is fast with no by-product; **step-growth** (condensation) is slow and releases a small molecule; **coordination** (Ziegler–Natta) gives neat stereoregular chains.
- **Thermoplastics** re-soften and remould; **thermosets** set permanently hard.
- Key plastics: **PVC**, **Teflon**, **Bakelite**, **Nylon-6,6**, **carbon fibres**; elastomers: **Buna-S** & **Buna-N**.
- **Conducting polymers** need a conjugated backbone + doping; **biodegradable polymers** (PGA, PLA) break down into CO_2, H_2O and biomass.

EXAM-ORIENTED QUESTIONS

Q1. Define polymer, monomer and degree of polymerization. *[3 marks]*

Ans. A polymer is a macromolecule made of many repeating units joined by covalent bonds. A monomer is the small molecule that repeats. Degree of polymerization (n) is the number of repeat units in a chain.

Q2. What is functionality? How does it decide the structure of a polymer? [3 marks]

Ans. Functionality is the number of reactive sites on a monomer. Functionality 2 gives a linear chain; functionality 3 or more gives a branched or cross-linked network.

Q3. Distinguish between addition and condensation polymerization. [5 marks]

Ans. Addition (chain-growth) uses C=C monomers, is fast and gives no by-product (e.g. polythene). Condensation (step-growth) uses bi-functional monomers, is slow and releases a small molecule like H_2O (e.g. Nylon-6,6).

Q4. Explain the free-radical mechanism of chain-growth polymerization. [5 marks]

Ans. Initiation: an initiator (benzoyl peroxide) forms free radicals that attack a monomer. Propagation: the radical adds monomer after monomer, growing the chain fast. Termination: two chains combine or disproportionate, stopping growth.

Q5. What is coordination polymerization? Name the catalyst and give one advantage. [3 marks]

Ans. It is polymerization using a Ziegler–Natta catalyst ($TiCl_4 + Al(C_2H_5)_3$). Advantage: it gives stereoregular (isotactic), linear, high-density polymers with controlled structure, at low pressure.

Q6. Differentiate thermoplastics and thermosetting plastics. [3 marks]

Ans. Thermoplastics have linear chains, soften on heating and can be remoulded (PVC, nylon). Thermosets have a 3-D cross-linked network, set permanently hard and cannot be remoulded (Bakelite, epoxy).

Q7. Give the monomer and two uses each of PVC and Teflon. [3 marks]

Ans. PVC: monomer vinyl chloride ($CH_2=CHCl$); uses — pipes, cable insulation. Teflon (PTFE): monomer tetrafluoroethylene ($CF_2=CF_2$); uses — non-stick cookware, gaskets/seals.

Q8. How is Nylon-6,6 made and why is it called a polyamide? [3 marks]

Ans. By condensation of adipic acid (6 C) and hexamethylenediamine (6 C). The units are joined by amide (–CO–NH–) linkages, so it is a polyamide.

Q9. What are elastomers? Compare Buna-S and Buna-N. [3 marks]

Ans. Elastomers are rubber-like polymers that stretch and recover. Buna-S (butadiene + styrene) is tough and used in tyres. Buna-N (butadiene + acrylonitrile) is oil- and heat-resistant and used in fuel hoses and oil seals.

Q10. How do conducting polymers conduct electricity? [5 marks]

Ans. They have a conjugated backbone (alternating double bonds) giving delocalized π electrons. Doping (e.g. I_2 or Na) adds or removes electrons, creating mobile charge carriers (polarons) that drift along the chain → current. Example: polyacetylene.

Q11. What are biodegradable polymers? Give two examples with uses. [3 marks]

Ans. Polymers broken down by micro-organisms into CO_2, H_2O and biomass. PGA — dissolvable surgical sutures; PLA — packaging, cutlery and 3-D printing filament.

Engineering Chemistry

UNIT – 5 : Instrumental Methods & Applications

A friendly, exam-ready tour of spectroscopy and chromatography — from the EM spectrum and Beer's law to UV-Vis, IR, and HPLC, with clear block diagrams.

Topics in this unit

1. Electromagnetic spectrum
2. Beer–Lambert law
3. UV–Visible spectroscopy
4. IR spectroscopy
5. Chromatography & classification
6. HPLC: principle & instrumentation

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. Introduction & the Electromagnetic Spectrum

Modern chemistry rarely stops at test-tube colour changes. To find out *what* a substance is and *how much* is present, we shine energy on it and watch how it responds. These are the *instrumental methods of analysis*, and most of them are forms of *spectroscopy*.

DEFINITION

Spectroscopy: the study of the interaction of matter with electromagnetic (EM) radiation. By measuring the radiation a sample absorbs or emits, we learn about its structure and concentration.

EM radiation travels as a wave, described by its wavelength λ, frequency ν and energy E. These are tied together by one small but powerful equation:

$$E = h\nu = hc / \lambda$$

h = Planck's constant (6.626×10^{-34} J·s), c = speed of light. Note: shorter λ → higher ν → higher energy E.

When we arrange *all* EM radiations in order of increasing wavelength (decreasing energy), we get the *electromagnetic spectrum*.

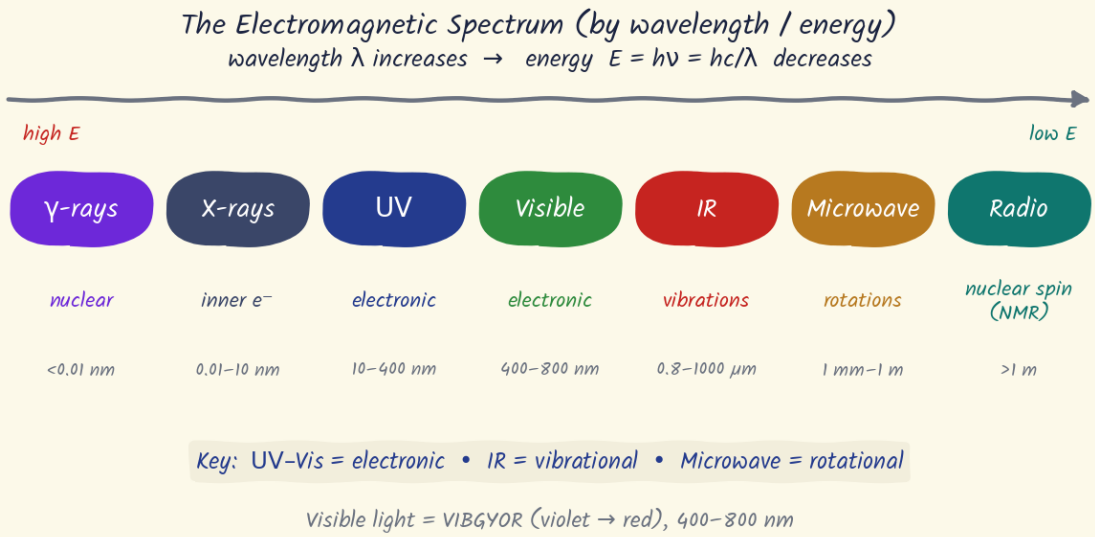


Fig 1 — The EM spectrum. Left → right: wavelength increases while energy decreases. Each region excites a different kind of change in matter.

- Order of regions: γ-rays → X-rays → UV → Visible → IR → Microwave → Radio.
- Moving right, wavelength increases and energy decreases ($E = hc/\lambda$).
- Visible light spans 400–800 nm (VIBGYOR: violet → red).
- The three regions we use most: UV-Visible → electronic transitions, IR → vibrations, Microwave → rotations.

> KEY POINT
> One idea unlocks the whole unit: each region of the spectrum carries a different amount of energy, so it excites a different motion in a molecule — electrons jump (UV-Vis), bonds vibrate (IR), molecules rotate (microwave).

2. Beer-Lambert Law

This is the single most important law for quantitative analysis — it tells us exactly how much light a solution soaks up. It combines two

simpler laws about a beam of monochromatic (single-wavelength) light passing through a solution.

DEFINITION

Lambert's law: the fraction of light absorbed is proportional to the path length (l) the light travels through the solution.

DEFINITION

Beer's law: the fraction of light absorbed is proportional to the concentration (c) of the absorbing solute.

Put together, they give the **Beer–Lambert law**:

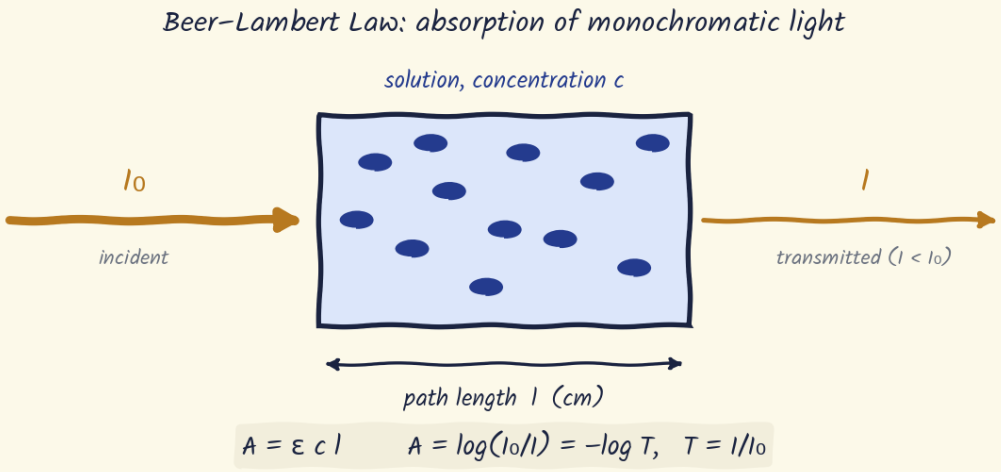


Fig 2 — Incident light I_0 enters a cuvette of path length l holding a solution of concentration c; weaker light I comes out.

$$A = \varepsilon c l$$

A = absorbance, ε = molar absorptivity (L mol^{-1} cm^{-1}), c = concentration (mol/L), l = path length (cm)

Absorbance and Transmittance

Two ways to describe the same measurement. **Transmittance (T)** is the fraction of light that gets through; **absorbance (A)** is how much is held

back.

$$T = I / I_0 \quad A = -\log T = \log (I_0 / I)$$

I_0 = incident intensity, I = transmitted intensity ($I < I_0$)

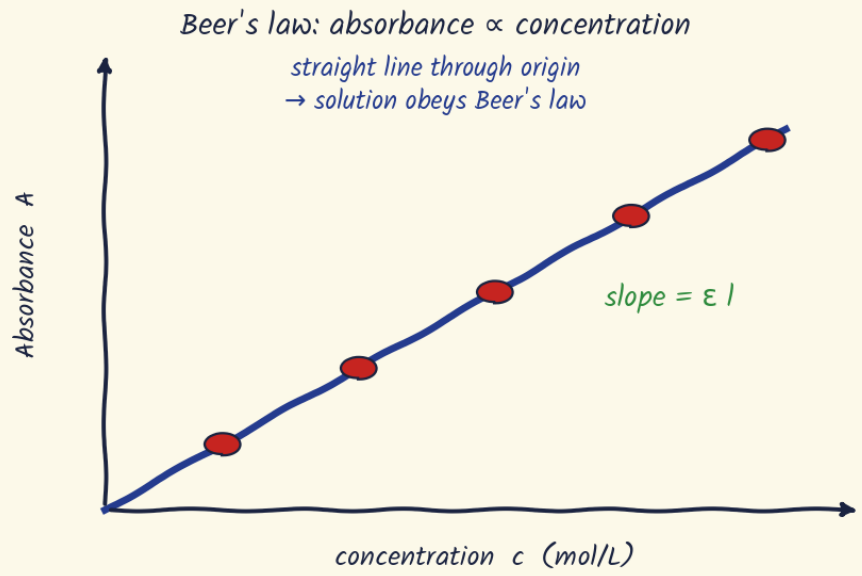


Fig 3 — Absorbance rises in a straight line with concentration (slope = εl). A line through the origin means the solution obeys Beer's law.

Limitations

- Holds only for dilute solutions (roughly $A < 1$); at high c the line bends.
- Needs truly monochromatic light.
- No chemical or physical change (association, dissociation, fluorescence) in the sample.

EXAMPLE

A 0.001 mol/L solution in a 1 cm cell has $\varepsilon = 2000$ L mol^{-1} cm^{-1}. Then $A = \varepsilon c l = 2000 \times 0.001 \times 1 = 2.0$ — doubling the concentration would double the absorbance.

3. UV-Visible Spectroscopy

Here we use UV light (200–400 nm) and visible light (400–800 nm). This radiation carries enough energy to push the molecule's valence electrons up to higher energy levels — so UV–Vis probes electronic transitions.

> DEFINITION
> **UV–Visible spectroscopy:** an absorption technique in which UV/visible radiation excites valence electrons from a lower to a higher molecular orbital. The wavelength absorbed tells us about the electronic structure.

Types of Electronic Transition

Electrons sit in σ, π and non-bonding (n) orbitals and can jump to the antibonding π* and σ* orbitals. The energy gap decides which wavelength is absorbed.

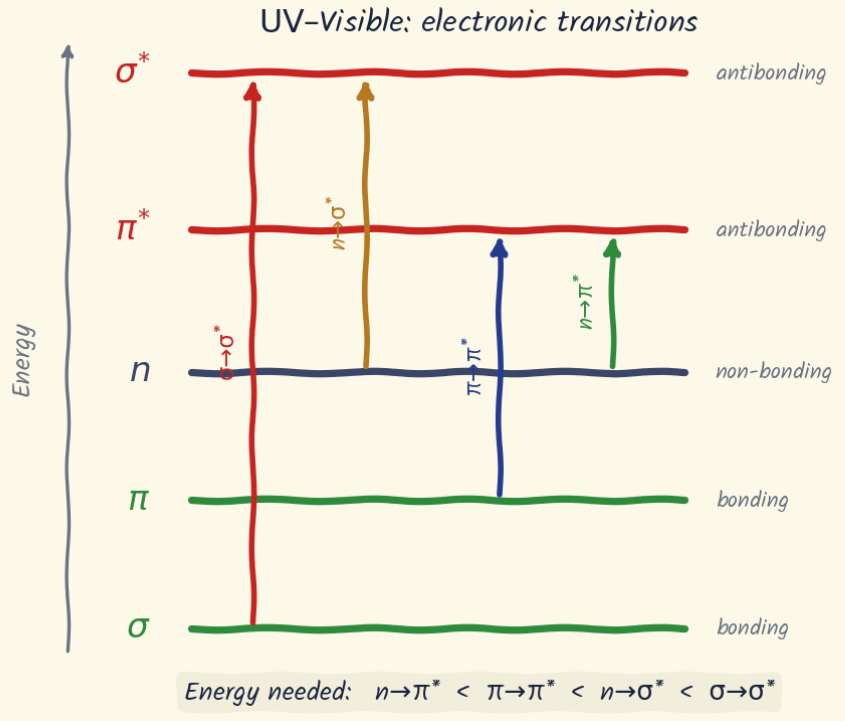


Fig 4 — The possible electronic transitions and their relative energies. n→π needs the least energy; σ→σ* needs the most.*

- $\sigma \rightarrow \sigma^*$ — highest energy (far UV); seen in saturated compounds like alkanes.
- $n \rightarrow \sigma^*$ — in molecules with lone pairs (e.g. alcohols, amines).
- $\pi \rightarrow \pi^*$ — in compounds with double bonds (alkenes, carbonyls).
- $n \rightarrow \pi^*$ — lowest energy; typical of C=O groups.
- Energy order: $n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$.

Chromophore & Auxochrome

DEFINITION

Chromophore: the group that actually absorbs UV-Vis light (the colour-giving unit). Examples: C=C, C=O, N=N and the benzene ring.

DEFINITION

Auxochrome: a saturated group with lone pairs (–OH, $-NH_2$, –Cl) that itself does not absorb, but when attached to a chromophore shifts and intensifies its absorption.

- Bathochromic (red) shift — absorption moves to longer λ.
- Hypsochromic (blue) shift — absorption moves to shorter λ.

Instrumentation

A UV-Visible spectrophotometer passes a single chosen wavelength through the sample and measures how much comes out, following the sequence below.

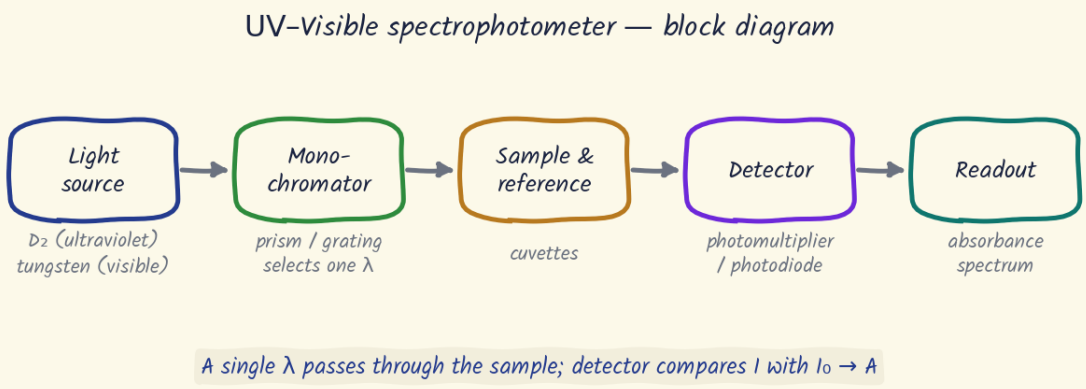


Fig 5 — Block diagram of a UV-Visible spectrophotometer.

- **Source** — deuterium (D_2) lamp for UV, tungsten lamp for visible.
- **Monochromator** — prism or grating that selects one wavelength λ.
- **Sample & reference** — matched cuvettes (sample vs blank).
- **Detector** — photomultiplier tube or photodiode converts light to current.
- **Readout** — records absorbance vs wavelength (the spectrum).

Applications

- **Quantitative estimation** of a substance using Beer's law ($A = \varepsilon c l$).
- **Detecting conjugation** — more conjugated double bonds absorb at longer λ.
- **Purity checking** and identification of unknown compounds.
- **Reaction kinetics** — following absorbance as a reaction proceeds.

> EXAMPLE
> $KMnO_4$ solution is purple because it absorbs green light (~525 nm). Its absorbance at that λ is directly proportional to concentration, so UV-Vis is used to measure how much $KMnO_4$ is present.

4. IR Spectroscopy

Infrared radiation carries less energy than visible light — not enough to move electrons, but just right to make bonds vibrate. So IR spectroscopy probes molecular vibrations and is the go-to tool for spotting functional groups.

DEFINITION

IR spectroscopy: an absorption technique using mid-IR radiation (4000–400 cm^{-1}, ≈ 2.5–25 μm) that excites the stretching and bending vibrations of covalent bonds.

KEY POINT

Selection rule: a vibration is IR-active only if it changes the dipole moment of the molecule. This is why symmetric molecules like O_2, N_2 and H_2 are IR-inactive — their stretch makes no dipole change.

Fundamental Modes of Vibration

Bonds move in two broad ways — the length can change (stretching) or the angle can change (bending / deformation).

IR vibrational modes of a 3-atom group (e.g. $-CH_2-$)

IR-active only if the vibration changes the dipole moment

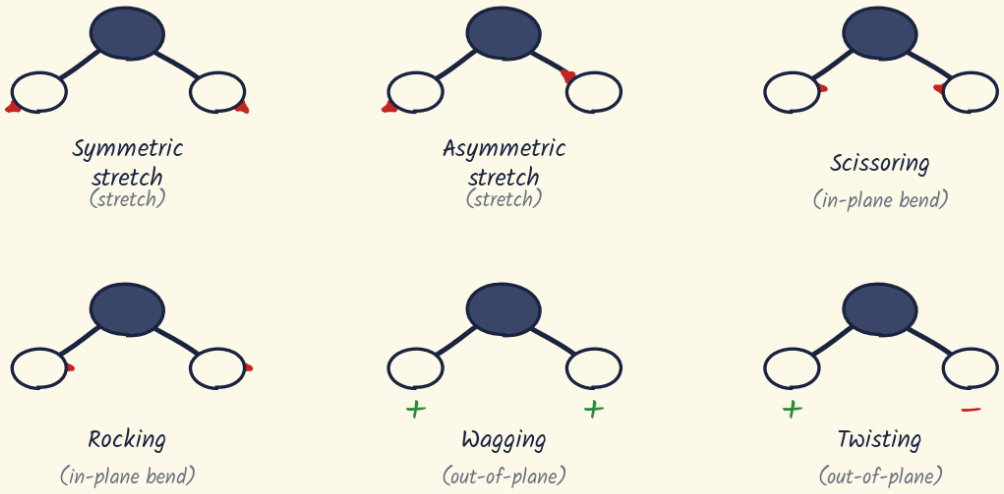


Fig 6 — Vibrational modes of a 3-atom group: two stretches and four bending modes (+ / − mean motion out of the plane).

- Stretching — symmetric and asymmetric (bond lengths change).
- Bending — scissoring, rocking (in-plane) and wagging, twisting (out-of-plane).

The number of fundamental vibrations depends on the number of atoms N:

Non-linear: $3N - 6$ Linear: $3N - 5$

e.g. H_2O (non-linear, N=3) → 3 modes; CO_2 (linear, N=3) → 4 modes

NOTE

The region below 1500 cm^{-1} is the fingerprint region — a complex pattern unique to each molecule, used to confirm identity (like a fingerprint).

Instrumentation

A dispersive IR spectrometer sends IR radiation through the sample and spreads it out to record how much is absorbed at each frequency.

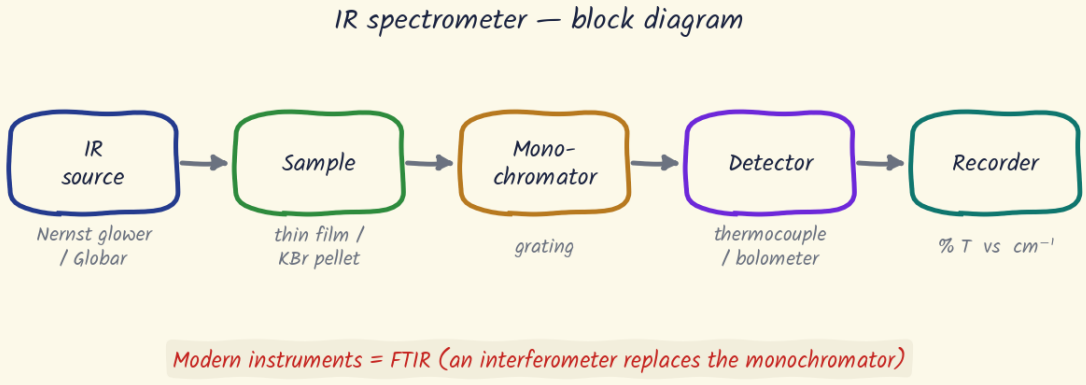


Fig 7 — Block diagram of an IR spectrometer.

- **Source** — Nernst glower or Globar (glows with IR when heated).
- **Sample** — as a thin film, liquid cell, or solid KBr pellet.
- **Monochromator** — a grating that disperses the IR beam.
- **Detector** — thermocouple or bolometer (senses heat).
- **Recorder** — plots % transmittance against wavenumber (cm^{-1}).

> NOTE
> Modern instruments are **FTIR** (Fourier-Transform IR): an **interferometer** replaces the monochromator, giving faster, more sensitive scans.

Applications

- **Functional-group identification** — e.g. a broad band near 3300 cm^{-1} = O–H; a sharp band near 1700 cm^{-1} = C=O.
- **Structure confirmation** of an unknown using the fingerprint region.
- Checking reaction progress and detecting impurities.

EXAMPLE

An alcohol and a ketone are easy to tell apart by IR: the alcohol shows a broad O–H stretch (~3300 cm^{-1}), while the ketone shows a strong C=O stretch (~1715 cm^{-1}).

5. Chromatography

The spectroscopic methods above identify and measure substances. Chromatography does something different — it separates a mixture into its pure components, so each can then be studied on its own.

DEFINITION

Chromatography: a technique that separates the components of a mixture based on their differential distribution between two phases — a fixed stationary phase and a moving mobile phase.

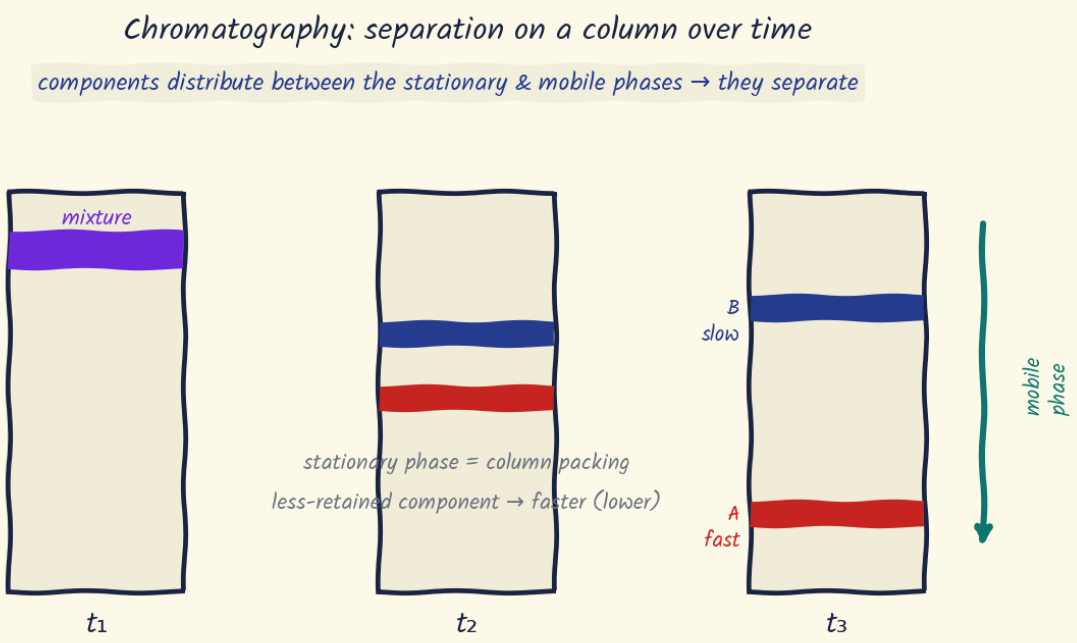


Fig 8 — As the mobile phase flows, each component spends a different time stuck to the stationary phase, so they travel at different speeds and separate.

- Stationary phase — stays fixed (paper, silica, a packed column).

- Mobile phase — flows and carries the sample (a liquid or gas).
- A component that clings more to the stationary phase moves slower; one that prefers the mobile phase moves faster → they separate.

$$R_f = \text{distance moved by solute} / \text{distance moved by solvent front}$$

The retardation factor R_f is constant for a compound under fixed conditions.

Classification

Chromatography can be grouped two ways — by the mechanism of separation, or by the technique used.

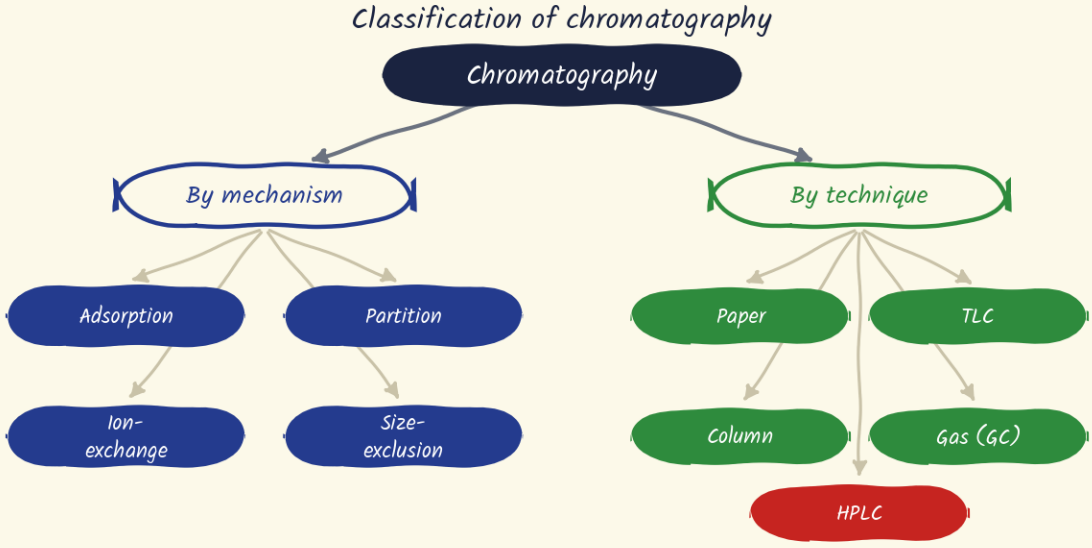


Fig 9 — Classification of chromatography by mechanism and by technique.

- By mechanism: adsorption, partition, ion-exchange, size-exclusion.
- By technique: paper, thin-layer (TLC), column, gas (GC) and HPLC.

EXAMPLE

In paper chromatography of ink, the paper is the stationary phase and the solvent is the mobile phase. The dyes separate into coloured spots, each with its own R_f value.

6. HPLC — High-Performance Liquid Chromatography

HPLC is column chromatography pushed to high performance. A liquid mobile phase is forced at high pressure through a column packed with very fine particles, giving fast, sharp and highly efficient separations.

DEFINITION

HPLC: a technique in which a liquid mobile phase is pumped under high pressure through a column of fine stationary-phase particles; the components separate by differential partition/adsorption and elute at different retention times, where a detector records them as peaks.

Instrumentation

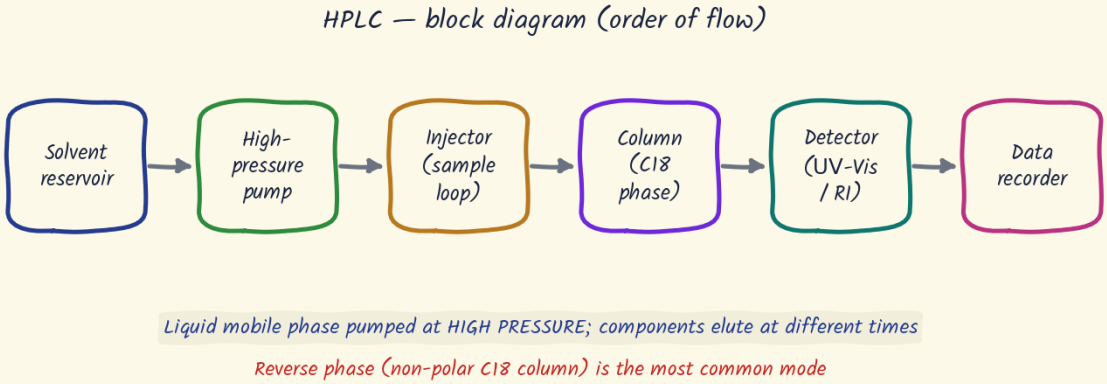


Fig 10 — Block diagram of an HPLC system, in the order the mobile phase flows.

1. Solvent reservoir — holds the mobile phase (eluent).
2. High-pressure pump — drives the solvent at constant high pressure.
3. Injector — introduces a fixed sample volume via a sample loop.
4. Column — packed with fine stationary-phase particles (the heart of HPLC).
5. Detector — UV-Vis, refractive-index (RI) or fluorescence, senses each component.

6. Data recorder — plots the chromatogram (peaks vs retention time).

Types of HPLC

Type	Stationary phase	Mobile phase	Note
Normal phase	polar (silica)	non-polar	polar compounds retained
Reverse phase	non-polar (C18)	polar	most common

Applications

- Separation and quantification of drugs (pharmaceutical analysis).
- Analysis of amino acids, proteins and biomolecules.
- Food and environmental testing (additives, pollutants).
- Purity testing and routine quality control (QC).

EXAMPLE

A pharma lab uses reverse-phase HPLC (C18 column) to check that a paracetamol tablet contains the correct amount of drug and no harmful impurities.

7. Applications & Importance

Instrumental methods are the backbone of modern analysis — fast, sensitive and needing only tiny samples:

- Pharmaceuticals: drug assay, purity and quality control by HPLC and UV–Vis.
- Structure determination: IR identifies functional groups; UV–Vis reveals conjugation.
- Environmental monitoring: detecting pollutants in water, air and soil.

- **Food & beverage:** checking additives, adulterants and nutrients.
- **Clinical & biochemical:** separating and measuring amino acids, proteins and metabolites.
- **Research & industry:** reaction monitoring, kinetics and material characterisation.

QUICK RECAP

- **Spectroscopy** studies how matter interacts with EM radiation; $E = h\nu = hc/\lambda$.
- EM spectrum order: $\gamma \rightarrow$ X-ray $\rightarrow$ UV $\rightarrow$ Visible $\rightarrow$ IR $\rightarrow$ Microwave $\rightarrow$ Radio (λ increases, E decreases).
- **Beer–Lambert law:** $A = \varepsilon cl$; $T = I/I_0$ and $A = -\log T$; valid for dilute solutions.
- **UV–Vis** excites valence electrons ($n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$); chromophores absorb, auxochromes shift absorption.
- **IR** excites vibrations (stretching & bending); IR-active only if the dipole moment changes; fingerprint region < 1500 cm^{-1}.
- **Chromatography** separates mixtures via a stationary + mobile phase; R_f = solute distance / solvent distance.
- **HPLC**: high-pressure liquid chromatography; reservoir $\rightarrow$ pump $\rightarrow$ injector $\rightarrow$ column $\rightarrow$ detector $\rightarrow$ recorder; reverse phase (C18) is most common.

EXAM-ORIENTED QUESTIONS

Q1. What is spectroscopy? Write the relation between energy, frequency and wavelength. [3 marks]

Ans. Spectroscopy is the study of the interaction of matter with electromagnetic radiation. Energy and radiation are related by $E = h\nu = hc/\lambda$, so shorter wavelength means higher energy.

Q2. Arrange the regions of the electromagnetic spectrum and state the change each causes. [3 marks]

Ans. In order of increasing wavelength: γ-rays → X-rays → UV → Visible → IR → Microwave → Radio. UV-Visible causes electronic transitions, IR causes molecular vibrations and microwave causes molecular rotations.

Q3. State and explain the Beer-Lambert law. Give its mathematical form. [5 marks]

Ans. Absorption of monochromatic light is proportional to both the path length (Lambert) and the concentration (Beer). Combined: $A = \varepsilon c l$, where A = absorbance, ε = molar absorptivity, c = concentration and l = path length.

Q4. Define absorbance and transmittance and relate them. [3 marks]

Ans. Transmittance $T = I/I_0$ is the fraction of light passing through. Absorbance A measures light held back: $A = -\log T = \log(I_0/I)$.

Q5. List the limitations of the Beer-Lambert law. [3 marks]

Ans. It holds only for dilute solutions, requires monochromatic light, and fails if the sample undergoes chemical or physical change (association, dissociation, fluorescence).

Q6. What are electronic transitions in UV-Visible spectroscopy? Give their energy order. [5 marks]

Ans. UV-Vis radiation excites valence electrons from σ, π or non-bonding (n) orbitals to antibonding π^* or σ^* orbitals. Energy order: $n\to\pi^* < \pi\to\pi^* < n\to\sigma^* < \sigma\to\sigma^*$.

Q7. Define chromophore and auxochrome with examples. [3 marks]

Ans. A chromophore is the group that absorbs UV-Vis light (C=C, C=O, N=N, benzene). An auxochrome (-OH, $-NH_2$, -Cl) does not absorb itself but shifts and intensifies the chromophore's absorption.

Q8. Draw/label the block diagram of a UV-Visible spectrophotometer. [5 marks]

Ans. Source (D_2 lamp for UV, tungsten for visible) → monochromator (prism/grating) → sample & reference cuvettes → detector (photomultiplier/photodiode) → readout.

Q9. When is a molecule IR-active? Why are O_2 and N_2 IR-inactive? [3 marks]

Ans. A vibration is IR-active only if it changes the dipole moment of the molecule. Homonuclear molecules like O_2 and N_2 have no dipole change on stretching, so they are IR-inactive.

Q10. Name the fundamental modes of molecular vibration studied in IR. [3 marks]

Ans. Stretching — symmetric and asymmetric; and bending — scissoring and rocking (in-plane), wagging and twisting (out-of-plane). Modes = $3N-6$ (non-linear) or $3N-5$ (linear).

Q11. What is chromatography? State its principle and classify it. [5 marks]

Ans. Chromatography separates a mixture by the differential distribution of its components between a stationary and a mobile phase. By mechanism: adsorption, partition, ion-exchange, size-exclusion. By technique: paper, TLC, column, GC, HPLC.

Q12. Explain the principle and instrumentation of HPLC and give two applications. [5 marks]

Ans. A liquid mobile phase is pumped at high pressure through a column of fine particles; components separate by differential partition and elute at different retention times. Set-up: reservoir → pump → injector → column → detector → recorder. Applications: drug analysis and protein/amino-acid separation.

Q13. What is a chromophore and an auxochrome? [3 marks]

Ans. A chromophore is the group that actually absorbs UV-Visible light (e.g. C=C, C=O, benzene). An auxochrome (e.g. –OH, $-NH_2$, –Cl) is an attached group that shifts the absorption to longer wavelength and makes it more intense.

Q14. List the main applications of UV-Visible and IR spectroscopy. [3 marks]

Ans. UV-Visible: quantitative estimation by Beer's law, detecting conjugation, purity checking and reaction kinetics. IR: identifying functional groups and confirming molecular structure using the fingerprint region.